KTH Industrial Engineering
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A cost-optimal and geospatial analysis for the power system of Sierra Leone

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KTH Industrial Engineering and Management	Master of Science Thesis A cost-optimal and geospatial analysis for the power system of Sierra Leone	
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Abstract

In 2014, the electricity access in Sierra Leone was almost 13.1%, consisting of 42% in urban areas and 1% in rural areas. The high transmission and distribution losses in the national grid, the insufficient generation capacity, and regulatory constraints are also some of the country´s challenges in the power sector (SEforALL Africa Hub, 2018). Nevertheless, the government of Sierra Leone has set a target to increase the electrification rate to 92% in 2030 (SEforALL Africa Hub, 2018). This target could be achieved by exploiting the abundant sources of renewable energy in such as hydro and solar.

The objective of this study is to analyze the investments in the power sector of Sierra Leone in order to cover the country´s future electricity needs considering national targets (electrification rate) and different tiers of electricity in the residential sector. The modelling tools, OnSSET, spatial electrification planning tool and OSeMOSYS, cost optimization for medium to long-run integrated assessment and energy planning tool are used for this thesis project.

In order to achieve future electricity target, the modelling period of this study has been set to 2015-2065. Under this study, three scenarios are analyzed, the reference, medium electricity demand, and high electricity demand for the period 2015-2065. In 2015, the consumption level was 578 kWh/household/year for the urban population and 73 kWh/household/ year for the rural population. The reference scenario considers Tier 3 (Global Tracking Framework, 2015) on electricity consumption for urban population and Tier 2 (Global Tracking Framework, 2015) for the rural population in 2065. The medium electricity demand scenario assumes slightly higher energy consumption (Tier 4 (Global Tracking Framework, 2015) for urban population and Tier 3 (Global Tracking Framework, 2015) for rural population) in 2065. Lastly, the high electricity demand scenario assumes the highest electricity demand (Tier 5 (Global Tracking Framework, 2015) for urban population and Tier 4 (Global Tracking Framework, 2015) for rural population) in 2065.

This study shows that for Sierra Leone, in order to cover its electricity needs in the future as well as to be fully electrified, its power generation will mainly be based on hydro. In the Reference scenario, where both OnSSET and OSeMOSYS analysis were used, it is suggested that 28% of the total electricity produced is to be generated by solar PV and 60% by hydropower plant. This is due to the fact that OnSSET, as a spatial analysis tool, also takes into consideration the distance between available resource and demand, on top of resource availability. Meanwhile, in the medium and high electricity demand scenarios, where only OSeMOSYS analysis was conducted, hydropower plant shows a more dominant contribution than the reference scenario. Around 68% of total electricity produced for medium electricity demand scenario is from hydropower plant. In the high electricity demand scenario, besides high electricity production from hydro (79.5% of total electricity produced), production from other technology such as HFO and solar PV is more evenly spread, especially in 2065.

Overall, it can still be deduced that hydro power plant is the most promising option for electricity generation in all scenarios. This is attributed mostly to its abundance as well as low production costs, such that even when the distance is considered, it is still reasonably more attractive than other options available.

Acknowledgements

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1. Introduction

The Energy system in Sierra Leone faces many challenges including insufficient generation capacity, a ruined transmission, and distribution system with very high technical and commercial losses, low voltage quality, and institutional and regulatory constraints (SEforALL Africa Hub, 2018). This unfortunate situation can be fixed by adjusting several factors in the electricity sector, such as: increasing financing/investment, enabling legal framework, improving distribution and transmission infrastructure, and improving quality awareness of solar PV and cooking stoves (Ababa, n.d.).

In 2016, the total population was 7,296,402. In 2030, the total population is predicted will be 10,038,361, and in 2065, the total population is predicted to be 19,694,584. The ratio of people living in a rural area is around 59%, and it is constant until 2065. The ratio between male and female in this country is almost equal, with 50,7% of the total population is female.

In 2016, the overall electricity access in Sierra Leone was around 20.3%, meaning 5,815,232 people did not receive electricity. From that electrification ratio, approximately 46.9% of the population had electricity access in the urban areas. On the contrary, in rural areas, only 2.5% of the population had access to electricity (SEforALL Africa Hub, 2018). Most of the electricity supply (90%) is concentrated in the four cities of Freetown, Makeni, Kenema, and Bo (Ashley-Edison, 2015). Due to an unstable electricity supply, petrol or diesel generator are operated frequently. Candles, Kerosene, and battery lamps are mainly used for lighting. However, there is compelling potential for the implementation of renewable energy, especially solar and hydroelectric power (UNDP, 2012).

Households/residents consume 62.5% of the total energy in Sierra Leone. However, in terms of electricity, household only consumes 37% of the total electricity, this happened due to the consumption of wood and charcoal for cooking, where household/residents consume more than 90% of the energy from this type of biomass. Meanwhile, industry/commercial consumes only 21% in terms of total energy and 60% of total electricity.

Table 1. Overall energy use in Sierra Leone (Ministry of Agriculture and Food Security (2012), PMU (2012), MEWR and NPA-BKPS (2012))

Sector	Biomass		Petroleum	Electricity (Grid Connected)	Total (%)
	Fuelwood	Charcoal			
Agriculture, Forestry, Fishing	1%	-	5%	2%	2%
Mining	-	-	9%	1%	2.5%
Industry/Commercial	3%	10%	12%	60%	21%
Transport	-	-	49%	-	12%
Household/Residents	96%	90%	25%	37%	62.5%
Total	100%	100%	100%	100%	100%

Sierra Leone has developed a SEforALL Action agenda in the framework led by ECREEE to reach 92% universal electricity access by 2030, which means there are still around 803,068 people have no access to electricity in that year. This agenda also envisages a sharp increase in renewable hydro production that will surpass 5000 GWh in 2030, from 150 GWh in 2013. It will also allow most of the electricity to be produced by hydro sources in 2030.

The objective of this thesis is to examine the power grid potential and financial investments required to support the electrification in Sierra Leone between 2015 and 2065. Renewable energy policy, electrification target, available resources, and other parameters are considered in this energy model. The Open Source energy MOdelling SYstem (OSeMOSYS) and Open Source Spatial Electrification Tool (OnSSET) are used to develop less costly electricity generation system in Sierra Leone. Three scenarios: reference, medium electricity demand, and high electricity demand scenario are implemented to compare the possibility of different electricity generation combination based on different availability of resources and different electricity targets.

2. Overview of Methodology

2.1 Methodology

Power sector model of Sierra Leone for the period of 2015–2065 is developed in both Open Source energy MOdelling SYstem (OSeMOSYS) and Open Source Spatial Electrification Tool (OnSSET).

- As a first step, OSeMOSYS is used to develop an electricity model to produce the least-cost power generation supply considering the resource availability in the country as well as the power generation capacity. However, the distance between power generation technology and the location of energy demand is not considered in this tool.
- As a second step, based on the results produced by the OSeMOSYS model, the Levelized Cost of Electricity is calculated, the value of which is used as a variable in the OnSSET analysis. Next, OnSSET will calculate the least costly solution between grid, mini grid, and stand-alone power generation supply by considering the distance between the technology and the location electricity demand.
- Lastly, the outputs of the OnSSET model are being used as inputs in the OSeMOSYS model. Then, the OSeMOSYS model is being run to identify more feasible power generation supply solution.

Power plants in Sierra Leone can be categorized into 5 generation types. These include hydro, biomass, Heavy Fuel Oil (HFO), Diesel, and PV power plants, where each category contains planned power plants, operated power plants, and potentially developed a power plant. Planned power plants are the ones which are currently planned and will be developed soon by the country, while operated power plants are power plants that are operated.

Electricity Demand from 2015 till 2030 is calculated based on population growth and electrification target by SEforALL Action agenda, and GDP for the other sector growth. However, household daily electricity consumption is assumed to remain stable. For the year 2031 till 2065, due to limited information about population forecast, electrification target is forecasted based on the previous year's population growth.

Three scenarios are developed in the model, which are the reference scenario, medium electricity demand scenario, and high electricity demand scenario. All scenarios have the same intermediate and final target, which is 92% electrification rate target in 2030 and 100% electrification rate target in 2065. Despite having the same target, each scenario has a different energy consumption level. The reference scenario assumes a Tier 3 electricity consumption level for urban population and Tier 2 for the rural population in 2065. The medium electricity demand scenario assumes higher electricity consumption level (Tier 4 for urban population and Tier 3 for rural population) in 2065. Meanwhile, the high electricity demand scenario assumes the highest electricity demand (Tier 5 for the urban population and Tier 4 for rural population) in 2065.

2.2 The Open Source Energy Modelling System (OSeMOSYS)

OSeMOSYS is an open-source energy modelling designed for medium to long-run energy planning. It is designed to fill the gap in the analytical toolbox available to the energy research community and energy planners in developing countries. This tool requires no upfront financial investment since it uses an open-source programming language (GNU MathProg) and uses the GNU Linear Programming Kit (GLPK) solver. Additionally, it requires a less significant learning curve and time commitment to build and operate. Those advantages allow communities of developers, academics, modellers, and even policymakers to use it.

The objective of OSeMOSYS is to calculate the lowest Net Present Cost of the energy system of a country to meet the energy demand (Howells, et al., 2011). The functionality of Osemosys can be relatively easily modified since it is being developed in a series of functional blocks, where each block is divided into different levels of abstraction. In addition, it is open-source, and it allows the energy analyst to develop and define the model complexity and to elaborate a wide range of applications.

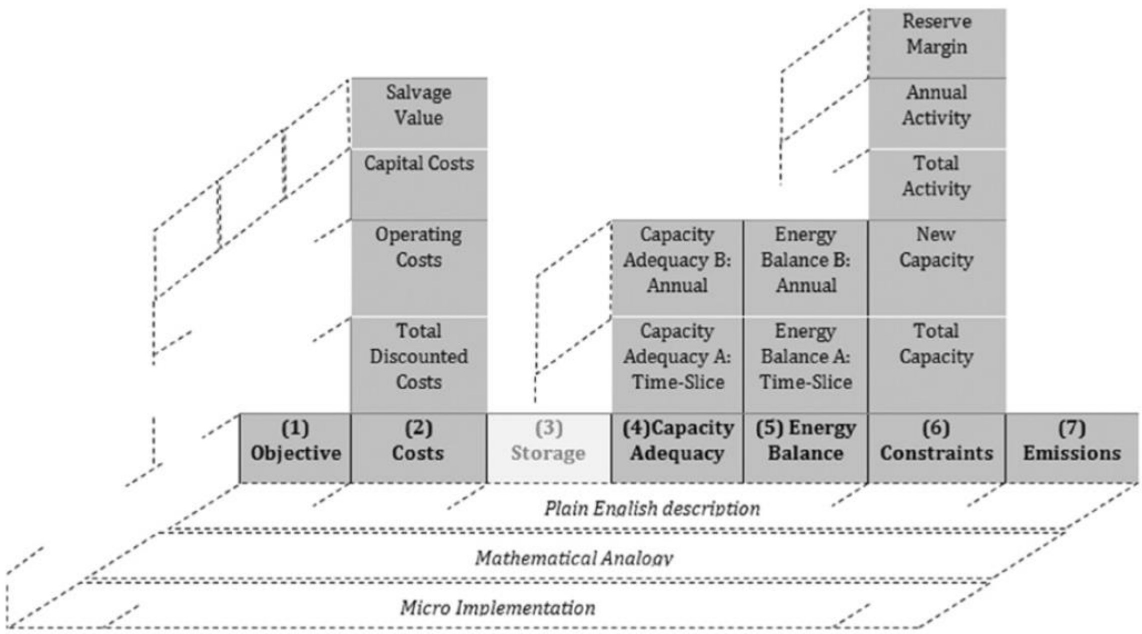


Figure 1. OSeMOSYS levels of abstraction.

Model Management Infrastructure (MoManI) is a browser-based open-source interface designed for energy systems modelling. It is developed to run the open energy modelling tool OSeMOSYS. MoManI helps the user to construct models, create several scenarios, and visualize results. MoManI has two main advantages. First, its unique structure allows multiple individuals or teams to cooperate simultaneously from all over the world. Second, even though developing an energy systems model could be a complex process, MoManI helps to simplify that process.

2.3 The Open Source Spatial Electrification Tool (OnSSET)

OnSSET is a GIS-based open source spatial electrification tool developed to identify the least-cost electrification option(s) between several alternative configurations. GIS analysis is needed to acquire a table that contains coordinates of each information such as demand, infrastructure, et cetera. The algorithm calculates the cost of electricity generation of each cell (particular area) for different electrification configuration based on the local specification and cost-related parameters of each specific cell. The calculated Levelized Cost of Energy (LCoE) for seven technology configurations represent the final cost electricity needed for the overall system. The calculated LCoE is then used to execute the extension algorithm. The modelling results will illustrate the technology mix, investment, and capacity required to achieve the electricity target of the modelled country within a specified period.

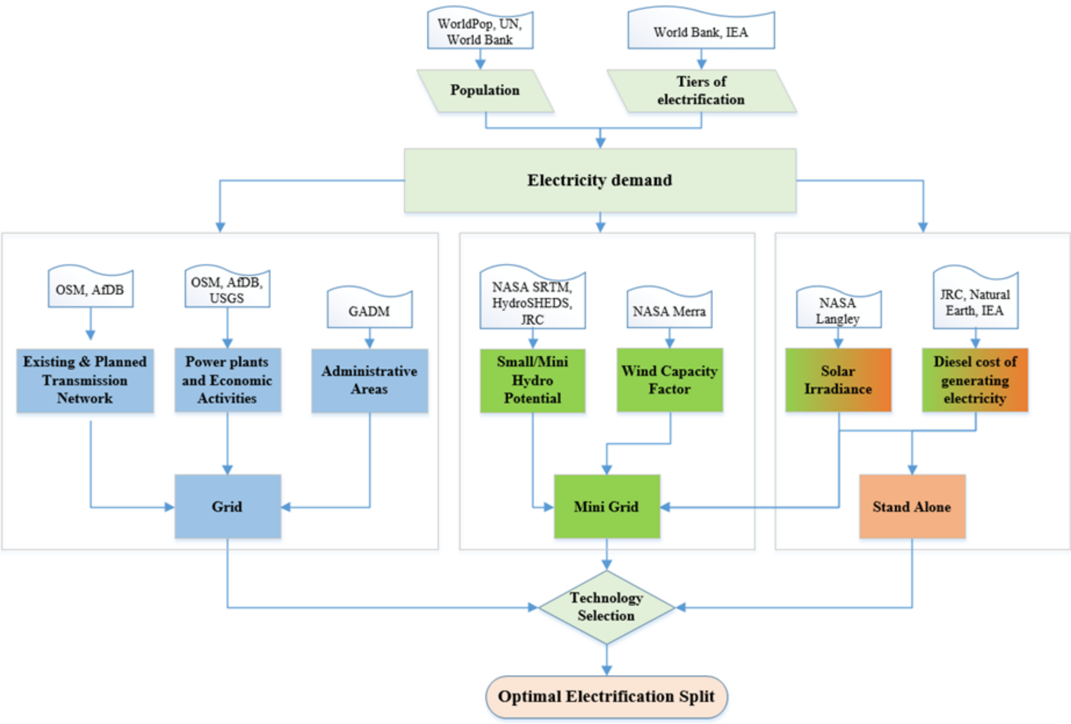


Figure 2. Methodology Overview of OnSSET

OnSSET requires QGIS and Anaconda packages to run. QGIS is a free and open-source GIS software. QGIS is needed to compile essential data for electrification analysis from GIS layers, combine them, and then extract them into a readable file in Python. QGIS is also useful to visualize modelling results in map form. Since OnSSET is written in Python, the Anaconda platform package, which contains various helpful Python modules, is required to run OnSSET successfully. Jupyter notebook (via Anaconda) is an interactive computing approach capable of encapsulating the whole computation process: developing, documenting, executing code, and representing results. It is used with OnSSET interface for analysis, exploring code and results.

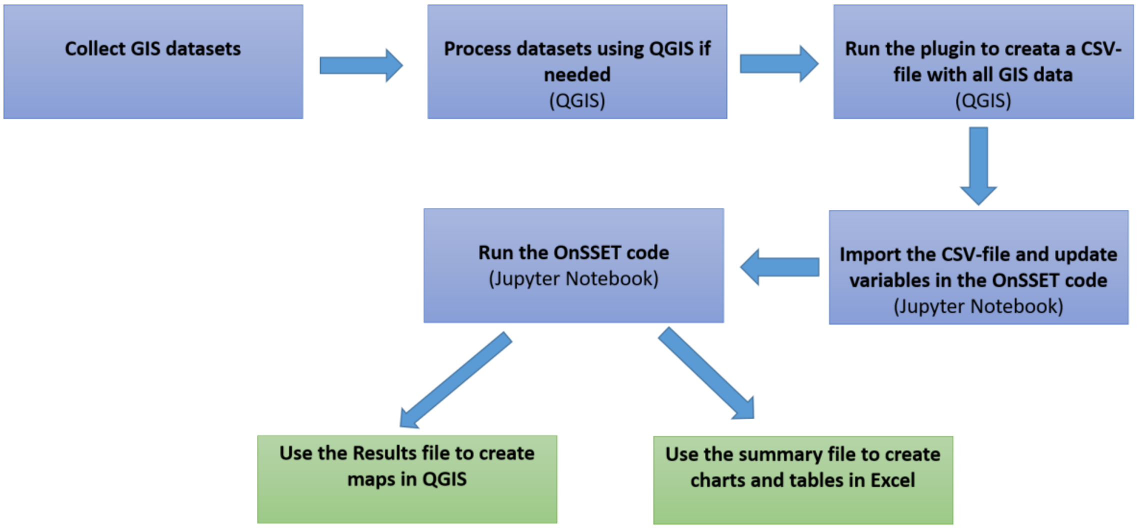


Figure 3. Necessary steps for OnSSET analysis

2.4 Reference Electricity System of Sierra Leone

Sierra Leone's Reference Electricity System is represented in the modelling framework in

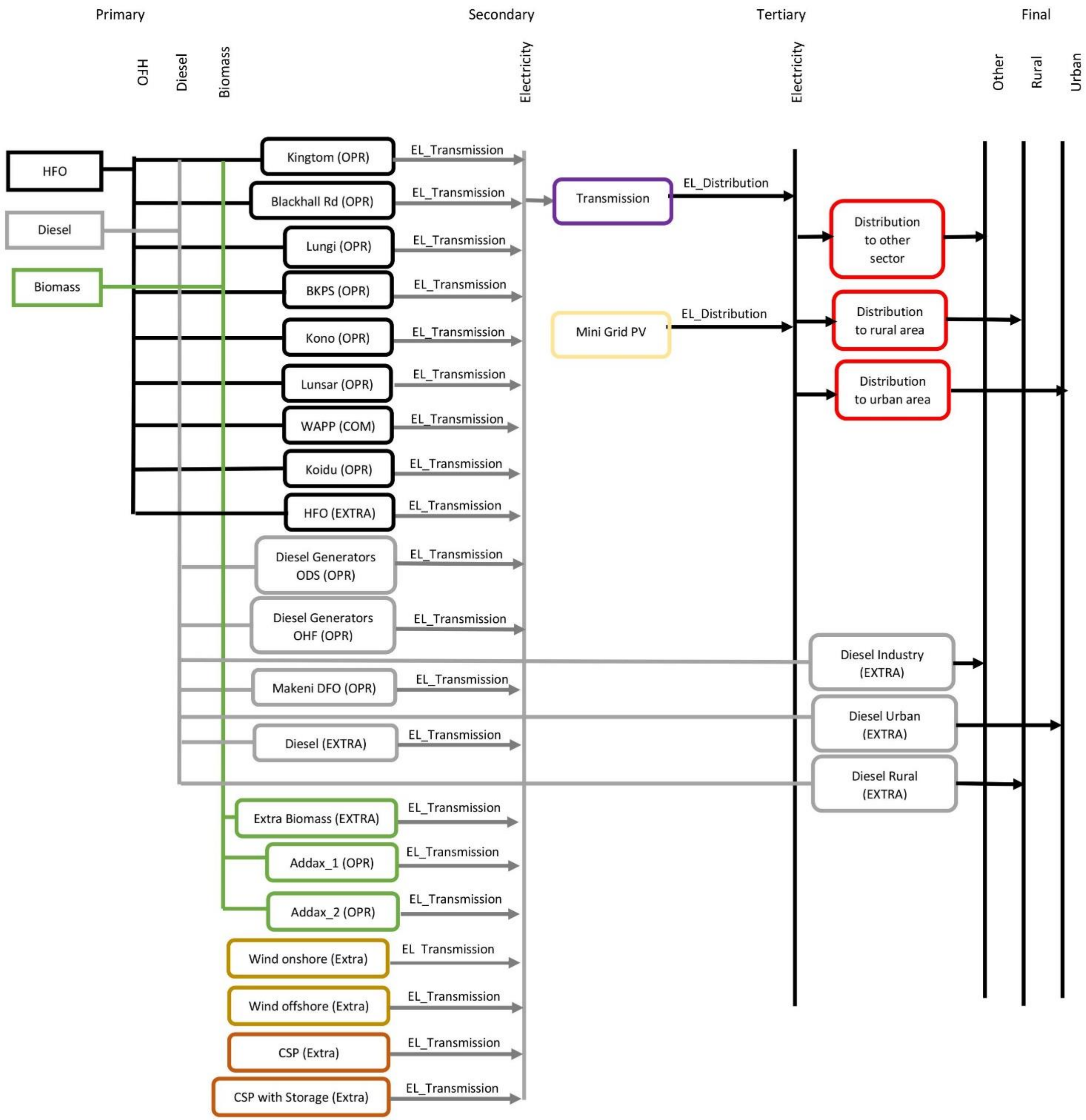
Figure 4. Technologies are being presented by boxes and lines for fuels. Within technologies, techno-economic parameters such as capacity, availability, lifetime, and cost are included.

In Table 1. , each power plant is represented with a different colour. Most of the power plants are connected to the transmission grid, while only two mini-hydro power plants, represented with dashed boxes, are connected directly to the distribution grid. On the final demand side, the electricity demand is divided into three main categories: urban population, rural population, and other electricity demand.

Extracted or imported fuel is defined as a primary energy, then it is transformed into electricity by each power plant. Electricity produced by these power plants is then transmitted through the transmission and distribution grid before finally distributed to each electricity consumer.

Table 2. The symbol for Sierra Leone power structure model structure

HFO	
Diesel	
Biomass	
Hydro	
Solar PV	
CSP	
Wind	


Primary
Secondary
Tertiary
Final
HFO
Diesel
Biomass
Electricity
Electricity
Other
Rural
Urban
HFO
Diesel
Biomass
Kingtom (OPR)
Blackhall Rd (OPR)
Lungi (OPR)
BKPS (OPR)
Kono (OPR)
Lunsar (OPR)
WAPP (COM)
Koidu (OPR)
HFO (EXTRA)
Diesel Generators ODS (OPR)
Diesel Generators OHF (OPR)
Makeni DFO (OPR)
Diesel (EXTRA)
Extra Biomass (EXTRA)
Addax_1 (OPR)
Addax_2 (OPR)
Wind onshore (Extra)
Wind offshore (Extra)
CSP (Extra)
CSP with Storage (Extra)
EL_Transmission
Transmission
EL_Distribution
Mini Grid PV
EL_Distribution
Distribution to other sector
Distribution to rural area
Distribution to urban area
Diesel Industry (EXTRA)
Diesel Urban (EXTRA)
Diesel Rural (EXTRA)

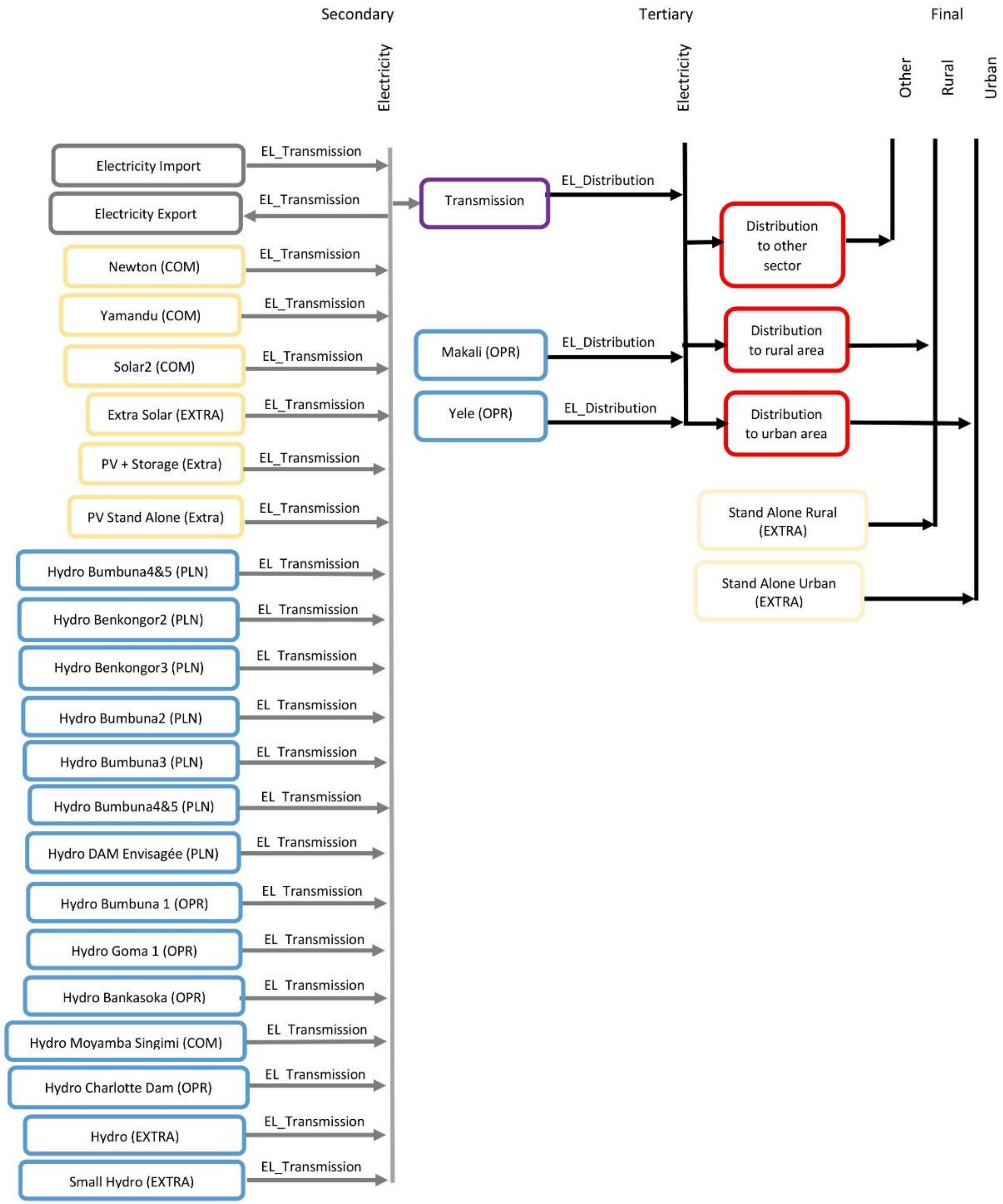


Figure 4. Sierra Leone power sector model structure

3. Scenario Description

Access to electricity is crucial to any country since it has impacts on a wide range of development indicators, including health, education, food security, gender equality, livelihoods, ad poverty reduction (The World Bank, 2018). Therefore, meticulous planning is essential to assist the country in achieving its electricity goals, especially for a developing country such as Sierra Leone. The development of scenarios could provide future pathways on electricity planning besides the uncertainties on fuel prices, electricity demands. Moreover, scenarios could provide a framework to analyze the various possibilities of technological development. Additionally, it will help illustrates how electricity system development will affect local or global issues of the country (Nakicenovic, et al., 2000).

In this study, three scenarios are developed to assess the possible outcome for different electricity consumption for the period of 2015-2065. All three scenarios contain the same committed electrification target set in SEforALL Action Agenda, which is 92% universal electricity access by 2030. By 2065, it is assumed that the electrification target will reach 100% in all three scenarios. Besides the electrification target, all three scenarios have same initial electricity consumption level, which is 578.327 kWh/household/year for urban household and 73 kWh/household/year for a rural household. Moreover, in 2030, it is assumed that the electricity consumption level has reached Tier 3 for the urban population and Tier 2 for Rural population for all scenarios.

- **Reference scenario**: it is developed based on Sierra Leone's current electricity situation and policy. Household electricity demand level in this scenario is assumed to reach Tier 3 for urban population and Tier 2 for the rural population in 2065, while another sector growth is expected to increase by 5.2% annually (The World Bank, 2019).

- **Medium electricity demand**: Sierra Leone's economy grew by 7.8% on average during 2003-2014 but contracted by 21% in 2015 following the Ebola outbreak and price decline of iron ore (The World Bank, 2019). In this scenario, it is assumed that Sierra Leone has recovered from the crisis and the other sector growth is stable at 7.8% since 2016. For the residential sector, in 2065, it is assumed that the urban population reaches Tier 4 in electricity consumption and Tier 3 for the rural population.

- **High electricity demand:** this scenario assumes the same other sector growth rates as in the medium electricity demand scenario. For the residential sector, in 2065, it is assumed that the urban population reaches Tier 5 in electricity consumption and Tier 4 for the rural population.

Table 3. Different level of electricity consumption per capita (Global Tracking Framework, 2015)

Level of Access	Consumption per household per year (kWh)
Tier 1	48
Tier 2	264
Tier 3	960
Tier 4	2538
Tier 5	3588

Table 4. Summary of the three different scenarios

Scenario	Electricity Consumption in 2030		Electricity Consumption in 2065		Other Sector Growth
	Urban	Rural	Urban	Rural	
Ref	Tier 3	Tier 2	Tier 3	Tier 2	5.2%
Med	Tier 3	Tier 2	Tier 4	Tier 3	7.8%
High	Tier 3	Tier 2	Tier 5	Tier 4	7.8%

4. Model Assumptions

4.1 Overall Assumptions

Overall assumptions across all scenarios and both OSeMOSYS and OnSSET are as follows:

- The discount rate chosen is 5% (Paul, 2011) because it is assumed that Sierra Leone currently in the economic development phase.
- Depreciation method value in the model is one, which means it has sinking fund depreciation.
- The model period is from 2015 to 2065.
- The monetary unit used is the 2015 USD rate.
- Assumptions on Transmission and Distribution losses (available in Table 12).
- Number of people per household in an urban and rural area: 6 (Statistics Sierra Leone and ICF International, 2014).
- Different levels of electricity access (Global Tracking Framework, 2015) (available in Table 3).
- The total number of time slices is 16; it is explained in more detail in Table 5.
- There are two-day types in a week, the first type is weekday which consists of 5 days in a week and the second type is weekend which has two days in a week

4.2 OSeMOSYS

The OSeMOSYS analysis is conducted to develop the three scenarios: reference scenario, medium electricity demand scenario, and high electricity demand scenario. Hence all necessary parameters will be considered to analyze all three scenarios, namely: electricity demand, power generation technologies, technology emission factors, fuel availability and prices, renewable resource potential, electricity generation options, and local transmission and distribution technologies.

4.2.1 Electricity Demand

In 2015, the total electricity demand was 279 GWh, the household sector consumed 38% of it, and the rest of it is consumed by other sectors, predominantly industries. In this study, the initial yearly electricity consumption for a household in the urban area has been assumed to be 578.32678 kWh, and 73 kWh in the rural areas, where each household consists of around six people. In 2030, the electricity consumption is set to Tier 3 for urban areas and Tier 2 for the rural area. Meanwhile, as discussed in section 3, the electrification target in 2030 and 2065 is set to 92% and 100% respectively.

The projected population in Sierra Leone from 2016 to 2030 is obtained from Statistics Sierra Leone report (SSL). While from 2031 to 2065, the value is projected linearly based on the growth rate in 2030.

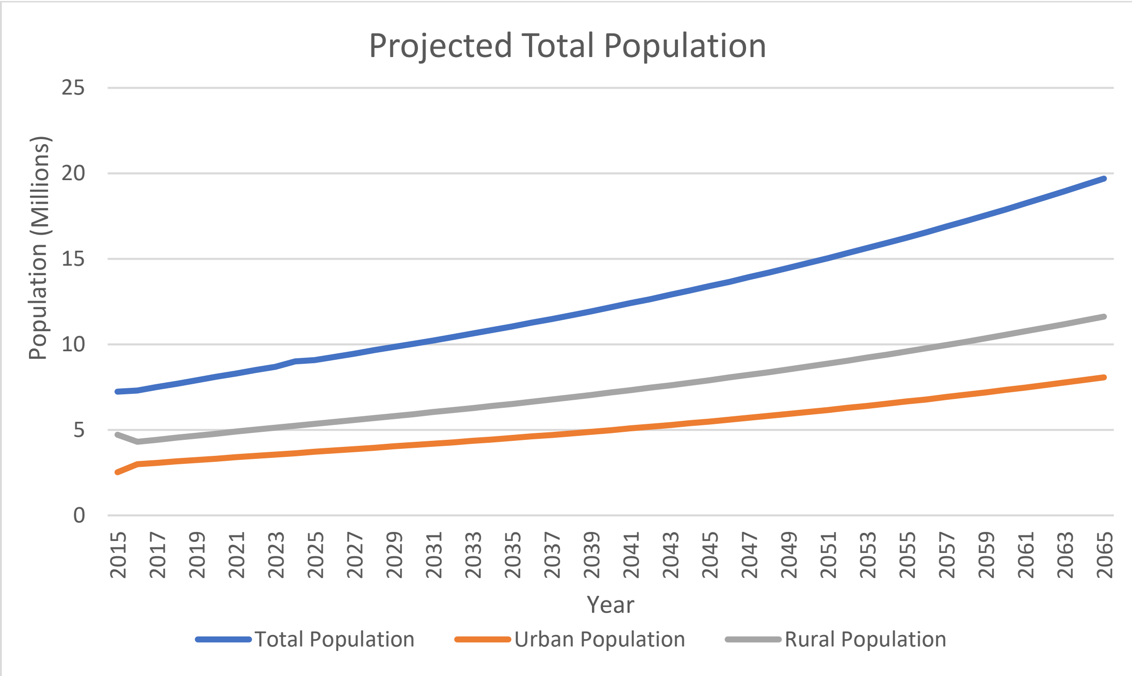


Figure 5. Projected population in Sierra Leone (Togoh, et al., 2017)

By knowing the electricity consumption, electrification rate target, and the projected number of populations, the future electricity demand for households can be calculated. Meanwhile, for other sectors, electricity projection is calculated based on the annual GDP growth rate.

The electricity demand was 1.0044 PJ in 2015. In 2030, the electrification rate is set to reach 92%, electricity consumption for the urban household is set to Tier 3 and rural household to Tier 2, and other sectors growth rate is set to 5.2%. Hence, total electricity demand is projected to be 4.32 PJ in 2030. Whereas, in 2065, where the electrification rate reaches 100% while maintaining the same tier of electricity consumption, the total electricity demand is expected to be 14.05 PJ.

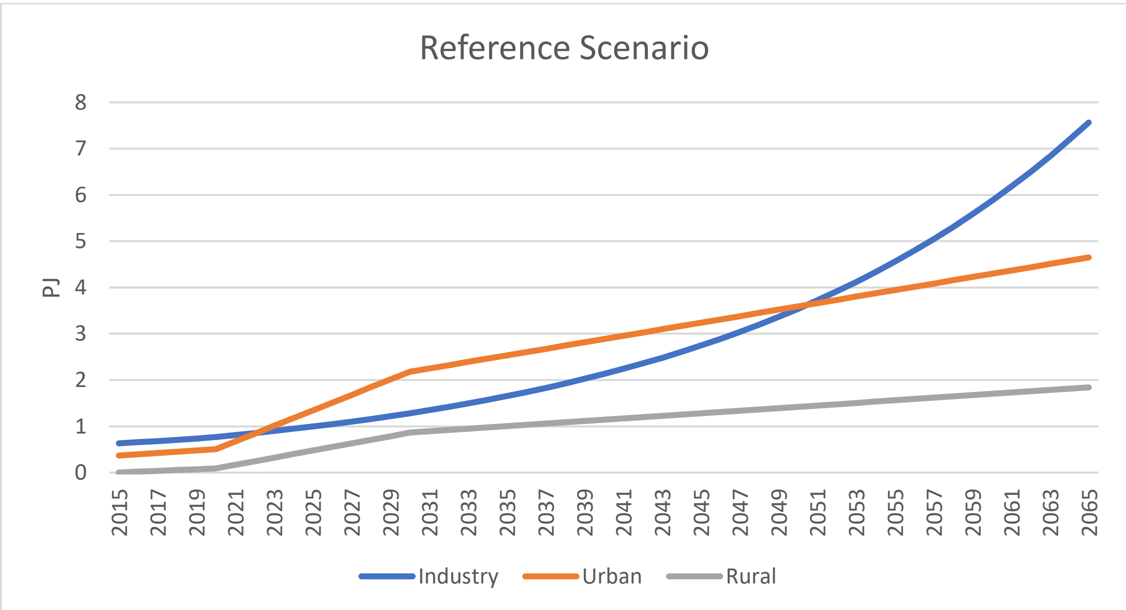


Figure 6. Electricity Demand in Reference Scenario

In the Medium Electricity Demand scenario, which the other sector growth rate is 7.8%, electricity Tier is set to 3 for urban household and 2 for rural household, the electricity demand is 4.99 PJ in 2030. In 2065, where the household electricity consumption is increased to Tier 4 for urban household and Tier 3 for rural household, and keeping the same others sector growth rate, electricity demand is predicted to be 46.0375 PJ.

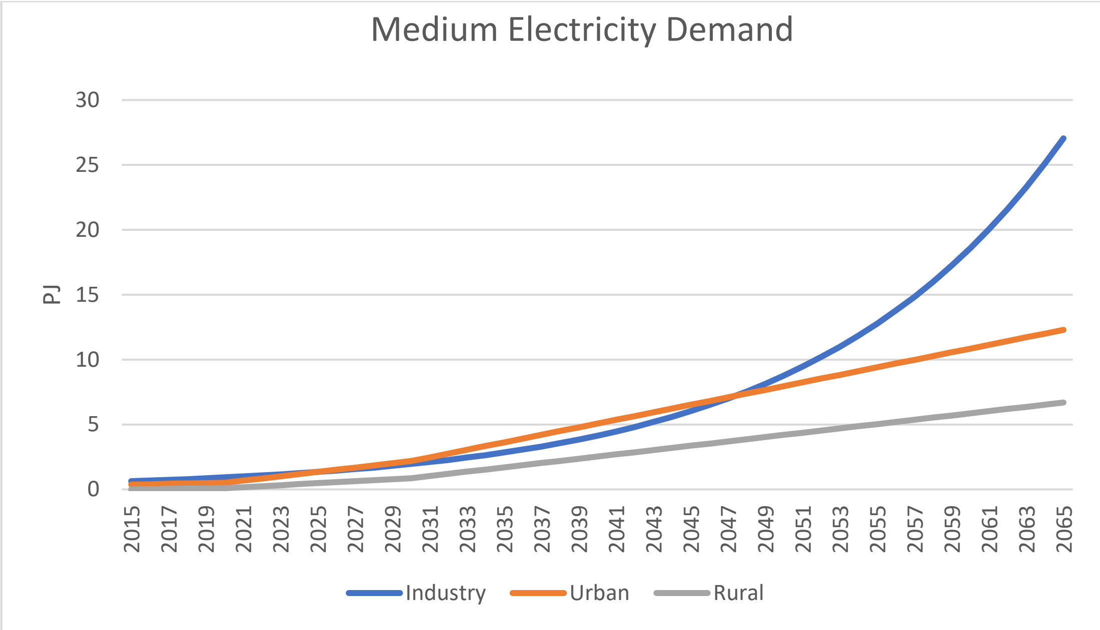


Figure 7. Electricity Demand in Medium Electricity Demand Scenario

In High Electricity Demand scenario, electricity demand in 2030 is similar to the value in the Medium Electricity demand scenario. However, by 2065, where the Tier of electricity is set to 5 for urban household and 4 for urban household, and having the same other sector growth, the electricity demand is 62.127 PJ.

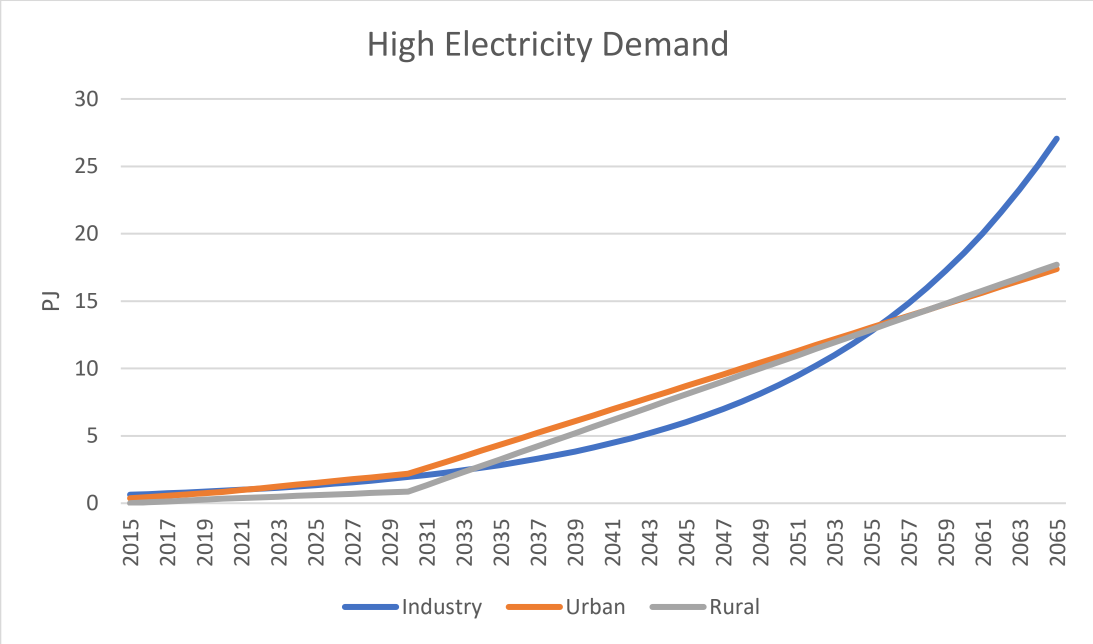


Figure 8. Electricity Demand in High Electricity Demand Scenario

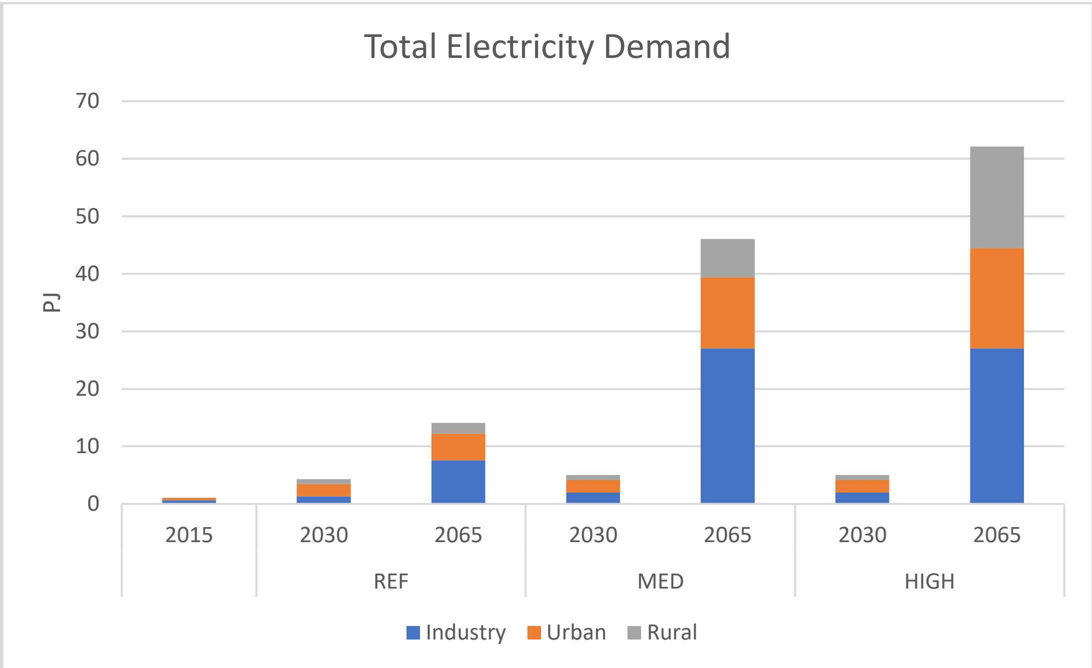


Figure 9. Comparison of different electricity consumption for the rural household sector

Electricity demand for each time slice can be determined based on the yearly load profile. Based on the graph, electricity consumption is almost constant throughout the year; however two peaks occur around April and October (KTH dESA, 2015). Weekly and the daily electricity consumption are analyzed for more detailed analysis. Overall, the daily electricity consumption for weekdays and weekends for a household in Sierra Leone is similar; however, on weekends (Saturday and Sunday), people tend to consume slightly more. As for day and night electricity consumption behaviour, people in Sierra Leone tend to consume more electricity during night-time.

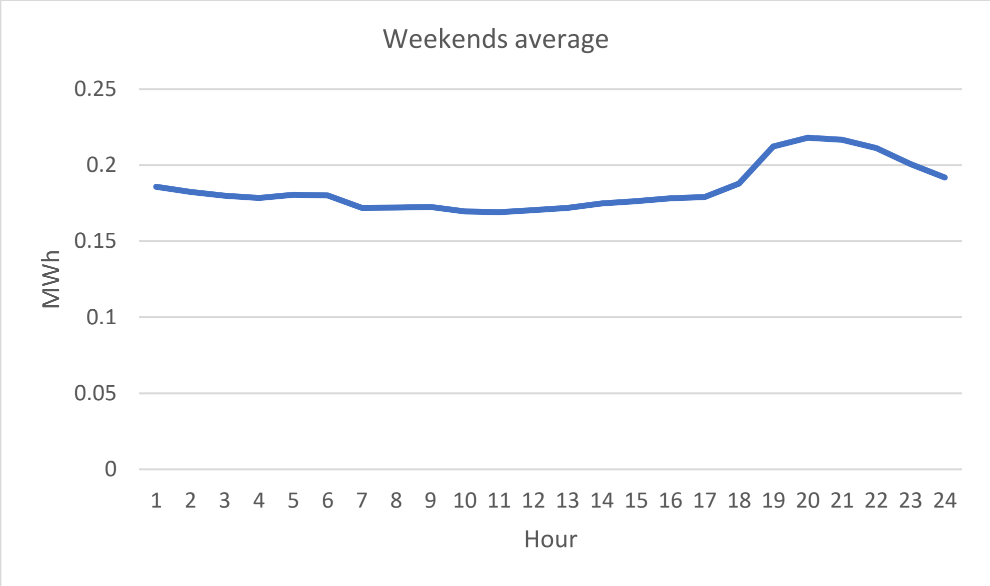


Figure 10. Daily electricity consumption on weekends (KTH dESA, 2015)

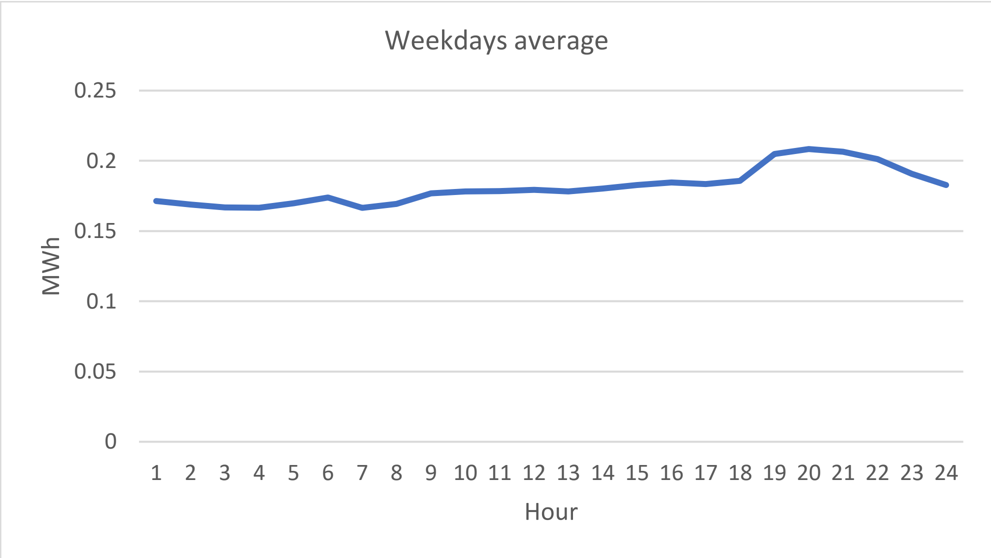


Figure 11. Daily electricity consumption on weekdays (KTH dESA, 2015)

Sierra Leone has two seasons, wet and dry season. The rainy season starts in May and ends in October, while the dry season is between November and April (National Tourist Board, 2017). These two distinct seasons cause substantial differences in water inflows to rivers, resulting contrast electricity production from hydropower technology (UNDP, 2012). Based on these situations, 16 time-slices are developed for this model. The time-slices consists of 2-day types (weekday and weekend), 4 seasons (wet and low electricity consumption level, wet and high electricity consumption level, dry and high electricity consumption level, dry and low electricity consumption level), and 2 daily time bracket (day and night).

Table 5. A detailed explanation of year split calculation for specified demand profile

Time Slice	Description
DDD1	Happens during the dry season when the electricity consumption in the season is low, on the weekday between 6.00 -18.00
DDD2	Happens during the dry season when the electricity consumption in the season is high, on the weekday between 6.00 -18.00
DDE1	Happens during the dry season when the electricity consumption in the season is low, on the weekend between 6.00 -18.00
DDE2	Happens during the dry season when the electricity consumption in the season is high, on the weekend between 6.00 -18.00
DND1	Happens during the dry season when the electricity consumption in the season is low, on the weekday between 18.00 – 6.00
DND2	Happens during the dry season when the electricity consumption in the season is high, on the weekday between 18.00 – 6.00
DNE1	Happens during the dry season when the electricity consumption in the season is low, on the weekend between 18.00 – 6.00
DNE2	Happens during the dry season when the electricity consumption in the season is high, on the weekend between 18.00 – 6.00
WDD1	Happens during the wet season when the electricity consumption in the season is low, on the weekday between 6.00 -18.00
WDD2	Happens during the wet season when the electricity consumption in the season is high, on the weekday between 6.00 -18.00
WDE1	Happens during the wet season when the electricity consumption in the season is low, on the weekend between 6.00 -18.00
WDE2	Happens during the wet season when the electricity consumption in the season is high, on the weekend between 6.00 -18.00
WND1	Happens during the wet season when the electricity consumption in the season is low, on the weekday between 18.00-6.00
WND2	Happens during the wet season when the electricity consumption in the season is high, on the weekday between 18.00-6.00
WNE1	Happens during the wet season when the electricity consumption in the season is low, on the weekend between 18.00-6.00
WNE2	Happens during the wet season when the electricity consumption in the season is high, on the weekend between 18.00-6.00

Table 6. A brief explanation of year split calculation for specified demand profile

Time Slice	Season	Electricity Consumption	Day type	Daily time bracket	Year Split
DDD1	Dry	Low	Weekday	Day	0.057745
DDD2	Dry	High	Weekday	Day	0.060114
DDE1	Dry	Low	Weekend	Day	0.058755
DDE2	Dry	High	Weekend	Day	0.061165
DND1	Dry	Low	Weekday	Night	0.063681
DND2	Dry	High	Weekday	Night	0.066294
DNE1	Dry	Low	Weekend	Night	0.064794
DNE2	Dry	High	Weekend	Night	0.067452
WDD1	Wet	Low	Weekday	Day	0.057745
WDD2	Wet	High	Weekday	Day	0.060114
WDE1	Wet	Low	Weekend	Day	0.058755
WDE2	Wet	High	Weekend	Day	0.061165
WND1	Wet	Low	Weekday	Night	0.063681
WND2	Wet	High	Weekday	Night	0.066294
WNE1	Wet	Low	Weekend	Night	0.064794
WNE2	Wet	High	Weekend	Night	0.067452

4.2.2 Power Generation Technologies

Power generation option in Sierra Leone consists of some fossil-fueled power plant and renewable technology power plant. However, renewable still dominates as the power generation option, due to the abundant resource of hydro and solar. Here is the list of power plants included in the model.

- Small hydro power sector (capacity up to 30 MW)
- Large hydro power sector (capacity more than 30 MW)
- Biomass power plant
- Diesel power plant
- HFO power plant
- Solar PV on the grid
- Solar PV with storage
- Solar PV stand-alone home system
- CSP
- CSP with storage

- On-shore wind power plant
- Off-shore wind power plant

The capacity factor for wind and PV Sierra Leone is calculated by averaging the capacity factor of wind and PV in three (3) different coordinates obtained from renewables.ninja website (Pfenninger & Staffell, n.d.). Different techno-economic parameter of the technology is gathered mainly from IRENA report and Annual Technology Baseline (ATB) (IRENA, 2018) (NREL, 2018). More detailed information for each power plant available in Sierra Leone is available in Appendix A Table 19.

Table 7. Techno-economic parameters for power generating technologies (IRENA, 2018)

Technology	Capital Cost (USD/kW)	Fixed Cost (USD/kW)	Variable Cost (USD/kWh)	Life (Years)	Efficiency	Availability	Construction time (Years)	Capacity Factor
Diesel 1 kW system (urban/rural)	750	23	51.52	10	0.16	0.3	0	0.56
Diesel 100 kW system (industry)	730	21	51.52	20	0.35	0.8	0	0.56
Diesel centralised	1240	35	51.52	25	0.35	0.8	2	0.56
Heavy fuel oil	1480	44	23.76	25	0.35	0.8	2	0.56
Large hydropower	2660	79.8	0	30	-	0.9	4	0.64
Small hydropower	4045	124	0	30	-	0.9	2	0.64
Biomass	3917	82	5.76	30	0.38	0.5	4	0.56
Onshore wind	1550.16	95	0	25	-	1	2	0.09
Offshore wind	4258.01	95	0	25	-	1	2	0.068
Solar PV (utility)	1753.92	30	0	25	-	1	1	0.1548
Solar PV (rooftop)	3782.04	50	0	20	-	1	1	0.1548
PV with battery storage	5010.00	61	0	20	-	1	1	0.255
CSP no storage	4010	65	0	25	-	1	4	0.35
CSP with storage	5900	93	0	25	-	1	4	0.4

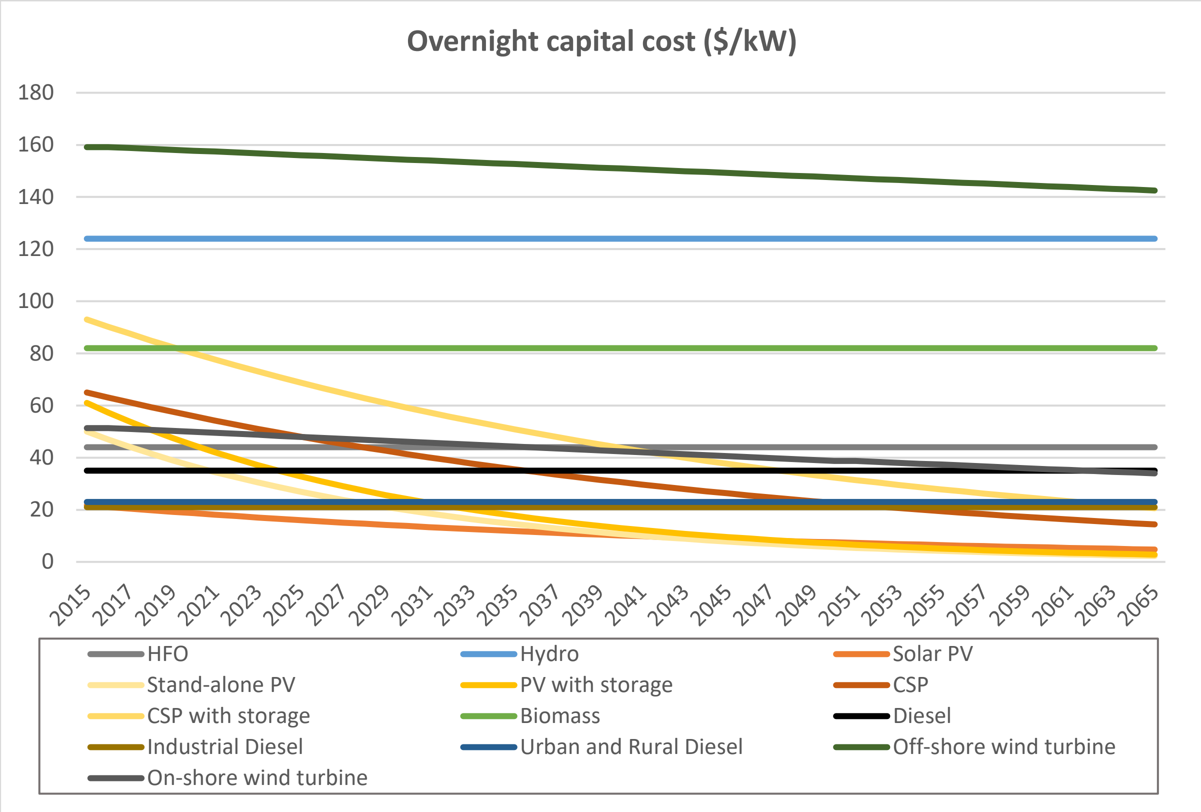


Figure 12. The overnight capital cost of each type of power plant

4.2.3 Technology Emission Factors

The thermal power plants emit carbon dioxide emissions which contributes to global warming. In order to capture the overall carbon dioxide emissions produced by the generated electricity, the following emission factors are assigned to the different type of fuels (Table 7). The emission factors are assigned in the import/extraction technologies.

Addax Bioenergy established a biomass-fuelled power plant using sugarcane ethanol as its fuel (REEEP, 2013). The production of electricity from sugarcane presented a low carbon footprint which is around 0.063 Mton CO_2/PJ (Junior, et al., 2019). Meanwhile for Diesel and HFO fuelled power plant, the carbon emission is higher than a biomass power plant, which is 0.0741 Mton CO_2 /PJ for diesel and 0.0774 Mton CO_2 /PJ for HFO (Verlag, 2019).

Table 8. Carbon dioxide emission factors per fuel

Fuel	Mton CO_2 /PJ
Diesel	0.0741
HFO	0.0774
Biomass	0.063

4.2.4 Fuel Availability and Prices

Sierra Leone's electricity production was mainly based on hydropower plants, out of a total 99.6 MW installed capacity in 2017 (Netherlands Enterprise Agency, 2018), Hydropower capacity is around 60 MW (IRENA, 2018). Besides hydro, Siera Leone also uses thermal power plants, such as biomass, diesel, and HFO power plant to cover its electricity needs. Biomass potential is high and includes fuelwood from forests, crop waste, and agricultural residue with an estimated total generation potential of 2,706 GWh (REEEP, 2013). However, Diesel and HFO fuels are mainly imported in the country (UN data, 2019).

Table 9. Price evolution for each fuel in Africa (IRENA, 2018)

USD/GJ	2015	2020	2030	2065
HFO (delivered to the coast)	6.6	10.2	14.3	31.2
HFO (delivered inland)	8.3	12.9	18.1	40.55
Diesel (delivered to the coast)	11.2	17.3	24.3	54.325
Diesel (delivered inland)	12.9	19.9	28.0	62.62
LCO (delivered to the coast)	9.1	14.1	19.8	44.32
Gas (domestic)	7.1	7.1	8.5	11.9
Gas (pipeline)	8.6	8.6	10.3	14.44
Gas (imported) [LNG]	9.2	8.7	11.0	15.54
Coal (domestic)	3.3	3.4	3.6	4.3
Coal (imported)	4.9	5.1	5.5	6.9
Biomass (moderate)	1.6	1.6	1.6	1.6
Biomass (scarce)	3.9	3.9	3.9	3.9

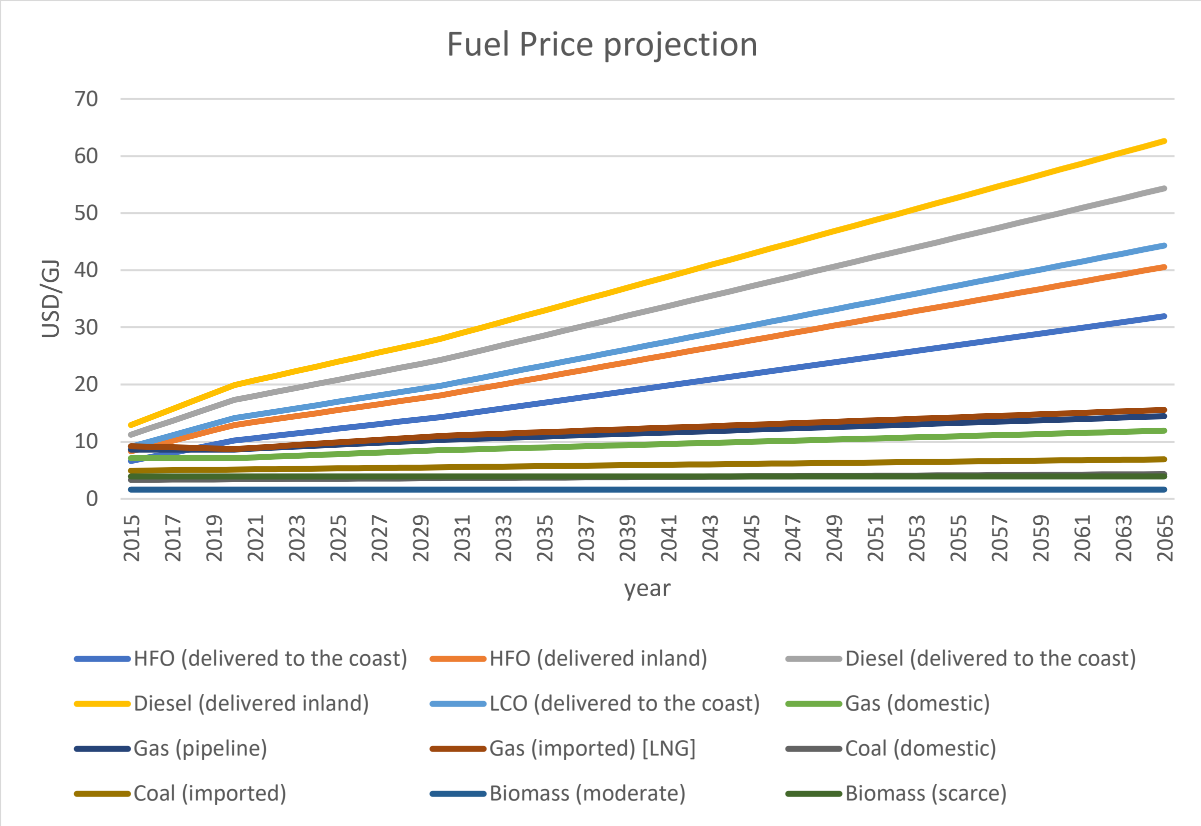


Figure 13. Fuel prices projection

4.2.5 Renewable Resource Potential

Sierra Leone has abundant renewable energy potential, primarily hydro and solar PV. Currently, the hydro power plants provide most of the electricity generation in Sierra Leone, with 56 MW hydro capacity from a total of 85 MW identified capacity in 2015. Moreover, it is identified that there is 5000 MW hydro. Now, Sierra Leone has planned several hydropower plant instalments ranging from 2 MW to 160 MW (Netherlands Enterprise Agency, 2018).

Even though the solar potential in the country is relatively high (1885 MW), the country has not invested in these type of technologies (solar PV, CSP). According to the "Renewable Energy Statistics 2017" report by IRENA, there is no solar power installed in Sierra Leone. However, in February 2017 Sierra Leone became the first country in Africa to sign the "Energy Africa Policy Compact" with the UK Government which contains a commitment to reach 250.000 households with modern energy solutions by 2018. Moreover, between 2014 and 2018, several initiatives such as Promoting Renewable Energy services for Social Development (PRESSD), Rural Renewable Energy Project, Installation of Solar Street Lights by Energy Ministry, and an expanded program by the Ministry of Health have enabled the implementation of solar mini-grids and solar-powered appliances in the country.

The maximum wind velocity in Sierra Leone averages between 3 m/s and 5 m/s, rising to around 8 m/s in mountainous areas. In specific locations, wind speed can reach about 12 m/s. Given these conditions, wind farms are possible in places such as along the coastline. With the availability of low wind speed turbines in the market, it is even more reason to install wind turbines in rural areas, especially in the north.

The government is exploring opportunities for developing small-scale biomass for rural electrification and the potential use of biodiesel from palm oil or ethanol for domestic consumption. Potential for a biomass power plant is high because of abundant forest resources and 656,400 tons of crop waste annually. Potential fuel for the power plants includes rice husks, straw, and sugar cane. Based on the available resources, it is estimated there is around 2,706 GWh generation potential.

Table 10. Estimates of technical potential for other renewable energy (IRENA, 2018)

Energy source	Hydro	Small hydro	Solar CSP	Solar PV	Biomass	Wind
Potential (MW)	5000	330	111	1885	587	131

4.2.6 Electricity Generation Options

Sierra Leone's power sector is relatively small, with less than 100 MW of operational capacity available and roughly 130.000 connected customers (USAID, 2016). Sierra Leone has a considerably high amount of renewable energy resources, it contributes to more than half of Sierra Leone's electricity production. Table 11 summarizes the existing power generating capacities by energy source in 2015.

Table 11. Existing power generating capacity as of 2015 (IRENA, 2018)

Power plant	Hydro	Oil	Solar	Biomass	Total
Capacity (MW)	56	21	0	8	85

As the population grows and the electrification rate target becomes higher, it is crucial to plan capacity investment carefully. Table 12 summarizes committed and planned power generation by energy source. In the model, only projects which are committed and currently running are considered, while the projects that are still under consideration, the optimization tool will decide if those are required to cover the country´s future electricity needs.

Table 12. Planned and committed power generating capacity additions (IRENA, 2018)

Power plant	Hydro	HFO	Solar	Biomass	Total
Planned Capacity (MW)	770.3	57	8	11	824
Committed Capacity (MW)	0	57	8	11	76

4.2.7 Local Transmission and Distribution Technologies

The country has suffered high electricity tariffs, which is among the highest in Africa at US$0.28 cents/kWh, and inadequate power capacity for many years. Moreover, electricity transmission and distribution has also been a significant challenge. In this study, transmission losses remain constant over time, while improvements in distribution technology help decrease distribution losses over time. Table 13. shows the evolution of transmission and distribution losses.

Table 13. Detailed transmission and distribution losses In Sierra Leone (IRENA, 2018)

	Cost (USD/kW)	Losses (%)			
		2015	2020	2030	2065
Transmission	120	5	5	5	5
Heavy industry	160	7	7	5	5
Urban residential/commercial	320	16–22.5	15	13	13
Rural residential	460	22.5–30	25	25	25

4.2.8 International Trade Links

Unpredictable increases in electricity demand incur the necessity to increase energy security to ensure uninterrupted access to electricity. Additionally, there may be problems which can cause blackouts in some parts of the country. Cross-border electricity trading can be one of the solutions to ensure a continuous supply of electricity. Sierra Leone has two neighbouring countries; Liberia and Guinea. These two countries provide possibilities for future electricity trading with Sierra Leone.

Guinea has abundant hydropower potential. Also, its location is geopolitically strategic. This strategic location provides an opportunity for Sierra Leone to conduct cross-border electricity trade with Guinea. Moreover, Guinea's ambition is to make use of low-cost hydroelectricity to become a significant exporter of electricity to neighbouring countries within the West African Power Pool. This potential further ensures the possibility for future electricity trade.

Meanwhile, Liberia's electricity condition, in general, is of the less fortunate situation compared to Sierra Leone, with only twelve percent total electrification rate. In the coming years, however, with plenty of renewable sources in Sierra Leona, and Liberia's electrification rate enhancement targets, opportunities for future electricity trading are open to possibilities.

Table 14. Detailed data for future cross-border transmission projects (IRENA, 2018)

From	To	Stations	Voltage kV	Capacity per line MW	Distance km	Losses %	Total investment USD million	Investment cost USD/kW	Earliest year
Liberia	Sierra Leone	Buchanan (LI) –	225	303.4	580	6.79 %	247.5	815.6	2018
Sierra Leone	Guinea	Monrovia	225	333.7	190	2.50 %	81.1	242.9	2018

4.3 OnSSET

The OnSSET analysis is conducted in the Reference scenario. Hence, only the parameters required to develop this scenario are considered.

4.3.1 Electricity Demand

Electricity demand considered in OnSSET analysis is only for rural and urban areas while other sectors such as industry are not considered. In 2015, urban electricity demand was around 0.3696 PJ, while rural electricity demand will be only just above 0.002 PJ. In 2065, the electricity demand for the urban area will be 4.65 PJ, while the rural will increase significantly to just above 1.841 PJ. A considerable increase in electricity demand is possible due to the high increase of electrification target from around 15% in 2015 while in 2065, the electrification target is set to 100%.

As in the assumption in OSeMOSYS, the intermediate target in 2030 is set to 92% electrification rate, and the final target in 2065 is set to 100%. The information about electrification situation and target is obtained from IRENA report, while population projection is obtained from Statistics Sierra Leone report. More detailed assumptions of demographic and social components for electricity demand calculation can be seen in Table 15.

Table 15. Demographics, social components, and electrification rate assumption

Variable	Value
Electrification ratio in 2015	0.153677686
Electrification ratio in 2030	0.92
Electrification ratio in 2065	1
Urban electrification ratio in 2015	0.422
Rural electrification ratio in 2015	0.01
Intermediate target year	2030
Urban population ratio in 2015	0.3487
Urban population ratio in 2030	0.41
Urban population ratio in 2065	0.41
The total population in 2015	7.237.025
The total population in 2030	10,038,361
The total population in 2065	19.694.584

4.3.2 Power Generation Technologies

Generally, improvement of electricity access has been achieved mainly through the extension of the national grid. However, grid extension is quite expensive and require a lot of time and commitment. On the contrary, a decentralized power system such as stand-alone or mini-grid power system can be a better solution to fulfil electricity demands in rural or remote areas. Moreover, in each remote area, there are usually resources available for mini-grid or stand-alone power generation that are capable of meeting electricity demands in those respective locations.

In OnSSET, there are seven electricity generation options included in the model. Those can be categorized into three main options: grid-extension, mini-grid, and stand-alone system.

Grid-extension

It may offer a lower power generation cost. However, it may not be feasible economically and socially if the demand occurs in remote areas.

Mini-grid - Wind Turbines, Solar PVs, Hydro, Diesel generators

Mini-grids usually utilize locally available energy resource or use diesel type power plant to provide electricity with several MW generating capacity. It offers a better solution than a grid extension for rural and remote areas with low or medium electricity consumption. In general, the mini-grids offer from moderate to high investment cost but the low operational cost to no fuel costs for renewable power plant. (Global Tracking Framework, 2015)

Stand-alone - Solar PVs, Diesel generators
Stand-alone systems are usually also based on available local energy resource or use diesel generator, but it often only produces few kWh of electricity daily. It is suitable to meet the electricity demand of a household or a small business. (Global Tracking Framework, 2015)

4.3.3 Electricity generation options

There are currently seven power generation technologies included in the model.

- Grid
- PV mini-grid
- Wind mini-grid
- Hydro mini-grid
- Diesel mini-grid
- PV Stand-alone systems
- Diesel Stand-alone systems

Electricity grid cost is obtained from OSeMOSYS analysis by calculating the LCOE of the grid power plants. The value calculated is 0.03345 USD/kWh. Meanwhile, information about capital costs necessary for the analysis can be found in Table 16.

Table 16. Electricity generation cost (IRENA, 2018)

Variable	Capital cost (USD/kW)
Stand-alone diesel	750
Mini-grid diesel	730
Mini-grid PV	758.27
Mini-grid wind	1413.14
Mini-grid hydro	3786.94
Stand-alone PV capital cost under 20 W	9620
Stand-alone PV capital cost between 21-50 W	8780
Stand-alone PV capital cost between 51-100 W	6380
Stand-alone PV capital cost between 101-1000 W	4470
Stand-alone PV capital cost over 1 kW	6950

4.3.4 Local transmission and distribution technologies

In OnSSET analysis, the distribution losses are considered to be 20% while the transmission losses to be 5% (IRENA, 2018). Variables necessary to calculate transmission and distribution can be seen in Table 17.

Table 17. Transmission and distribution-related cost (Mentis, et al., 2017) (Csanyi, 2011)

Variable	Value
HV line cost	53000 USD/km
MV line cost	9000 USD/km
LV line cost	3840 USD/km
HV to LV transformer cost	13500 USD/unit
HV to MV transformer cost	13500 USD/unit
MV to LV transformer cost	13500 USD/unit
MV to MV transformer cost	13500 USD/unit

5. Results

5.1 Reference scenario

The Reference scenario is modelled using both OnSSET and OSeMOSYS models. OnSSET is used to calculate the mini-grid and stand-alone power system considering the location of the demand because, in OSeMOSYS, the location of the electricity demand is not considered. The results of the OnSSET analysis can be found in Table 18. Those indicate the development of stand-alone and mini-grid PV, with a total of 282 MW new installation of this technology in 2030 and 402 MW in 2065. Besides, the model also suggests a small development of stand-alone diesel technology which the value is nearly zero.

Table 18. The capacity of the technology and population electrified with the specific technology model by OnSSET

Technology	Capacity 2030 (MW)	Population Electrified with specific technology in 2030	Capacity 2065 (MW)	Population Electrified with specific technology in 2065
Grid	40	1,541,474	95	7,482,885
Stand Alone PV	5	206,000	1	149,675
Mini Grid PV	277	6,275,298	401	12,036,886

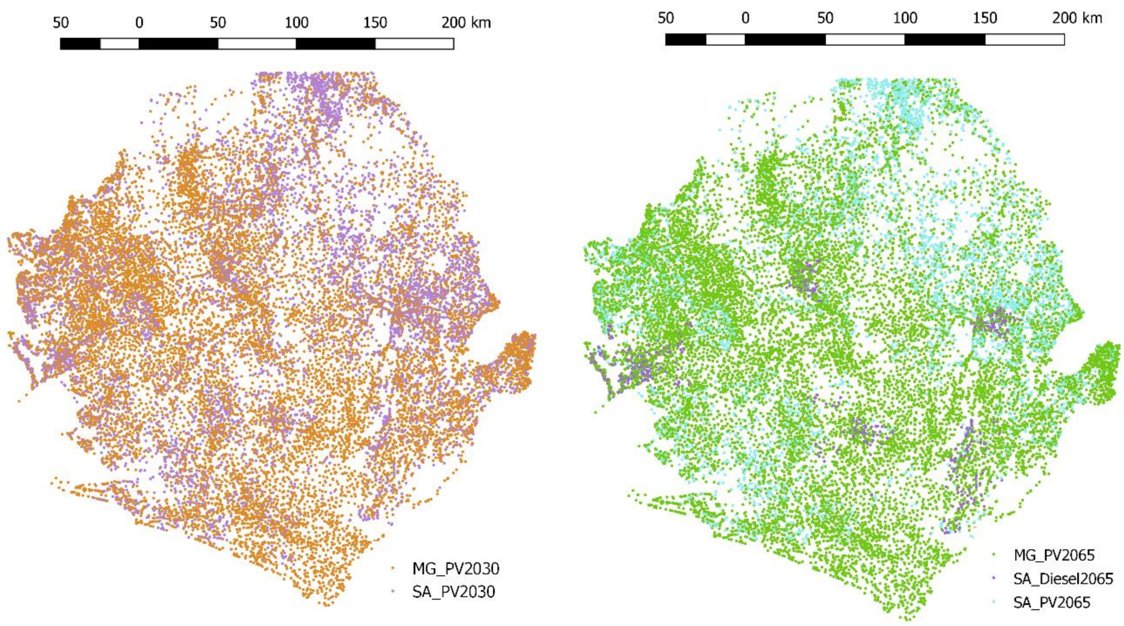


Figure 14. Off-grid development recommendation mapping by OnSSET in 2030 (left) and 2065 (right)

The potential development of each technology is widely spread across the country. Both stand-alone and mini-grid technologies are almost evenly spread throughout the country´s territory. Some capacity is denser in a specific area such as the west part of the country where the Freetown city as the most populated city is located. As electricity demand is increasing, and no efficiency measure is implemented, hence, the number of installed capacity is growing linearly following the demand.

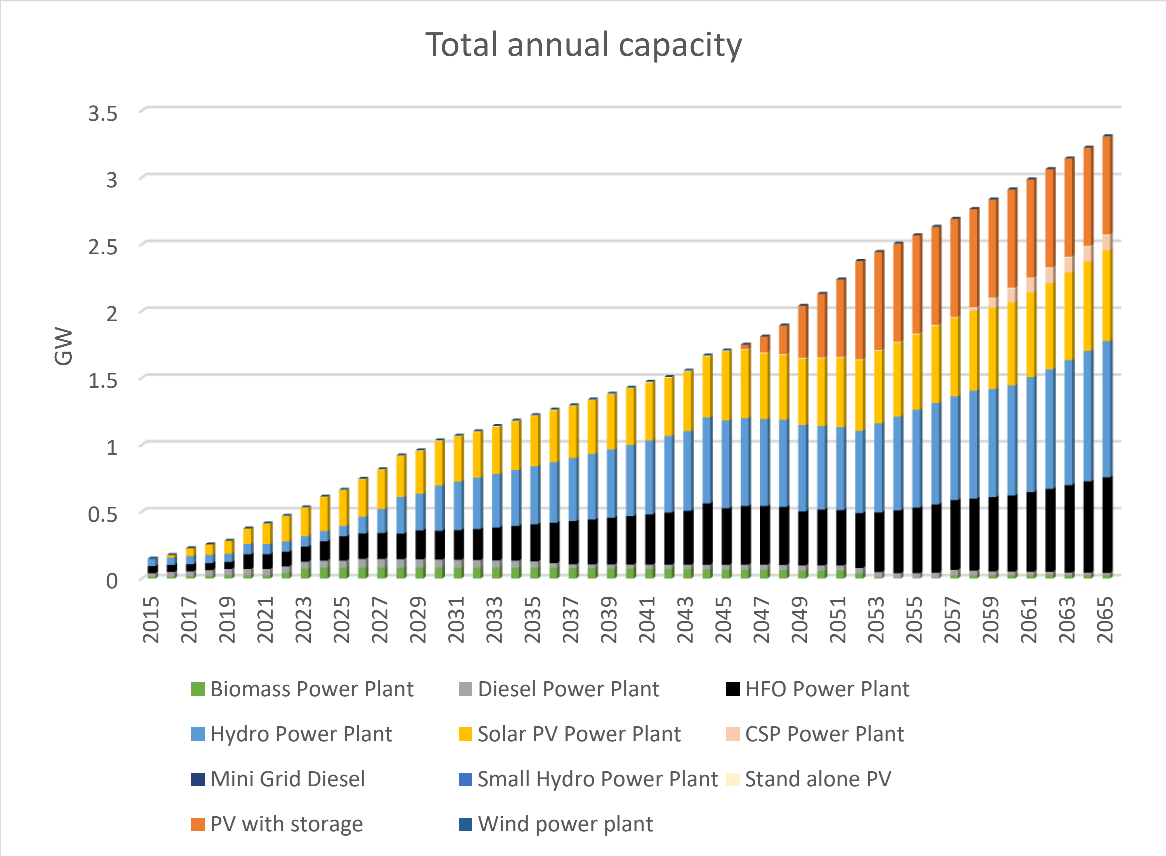


Figure 15. Total annual capacity by technology for the reference scenario

After the recommendation from OnSSET analysis is implemented in OSeMOSYS analysis, the result shows a total annual capacity was around 0.158 GW in 2015, while in 2030, it increases to 1.033 GW, and in 2065, to 3.3 GW. In 2015, hydro capacity was 0.056 GW, while in 2065, it increases to 1.02 GW. Solar PV investments start in 2016 (0.0383 GW) and the capacity increases by approximately 76.8 times between 2016 and 2065 for all kind of PV technology, including stand-alone PV, Medium Grid PV, and PV with storage. Biomass power plant capacity increased from 0.073 GW in 2015 to 0.196 GW in 2025, and gradually starts declining after 2035 until becoming zero in 2054. However, in 2056, biomass starts to produce electricity again.

Meanwhile, fossil-fueled power plant capacities show a steady contribution for the whole modelling period with an average annual capacity of 0.04 GW for Diesel power plant and 0.34 GW for HFO power plant. HFO technology shows a steady increase in capacity annually, with only 0.0523 GW capacity in 2015 and 0.715 GW in 2065.

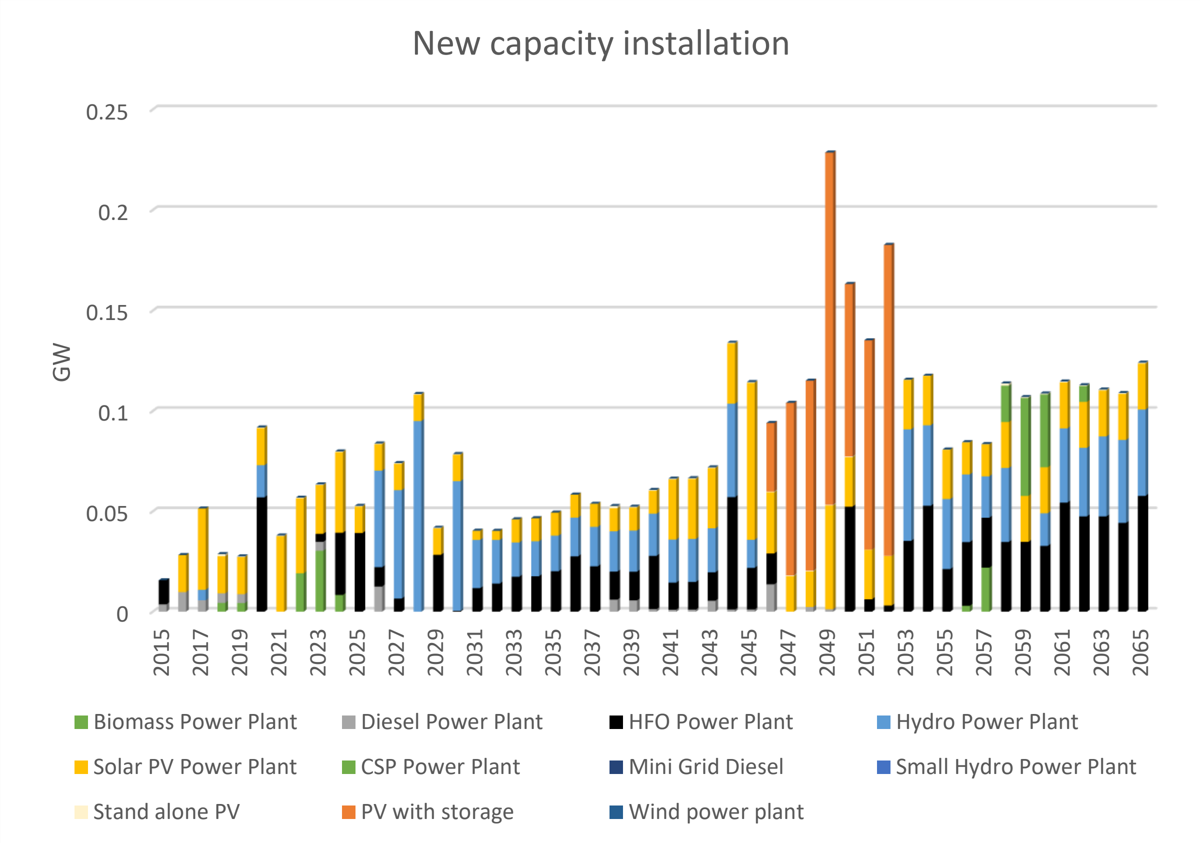


Figure 16. Annual capacity installation by technology for the reference scenario

The capacity of all PV technologies is continually increasing throughout the modelling period due to constant annual investment. Stand-alone solar PV is installed annually between 2016 until 2065, with an average annual investment of 0.001181 GW. Besides, medium grid PV technology is installed annually between 2016 until 2060, with an average 0.043 GW annual investment. Meanwhile, investments in solar PV with storage starts in 2049 with 0.075 GW and increases to its maximum potential (0.9315 GW) in 2056. The highest annual capacity investment happens in 2049 when the investment of PV with storage happens with a capacity investment close to three times of the average annual investment.

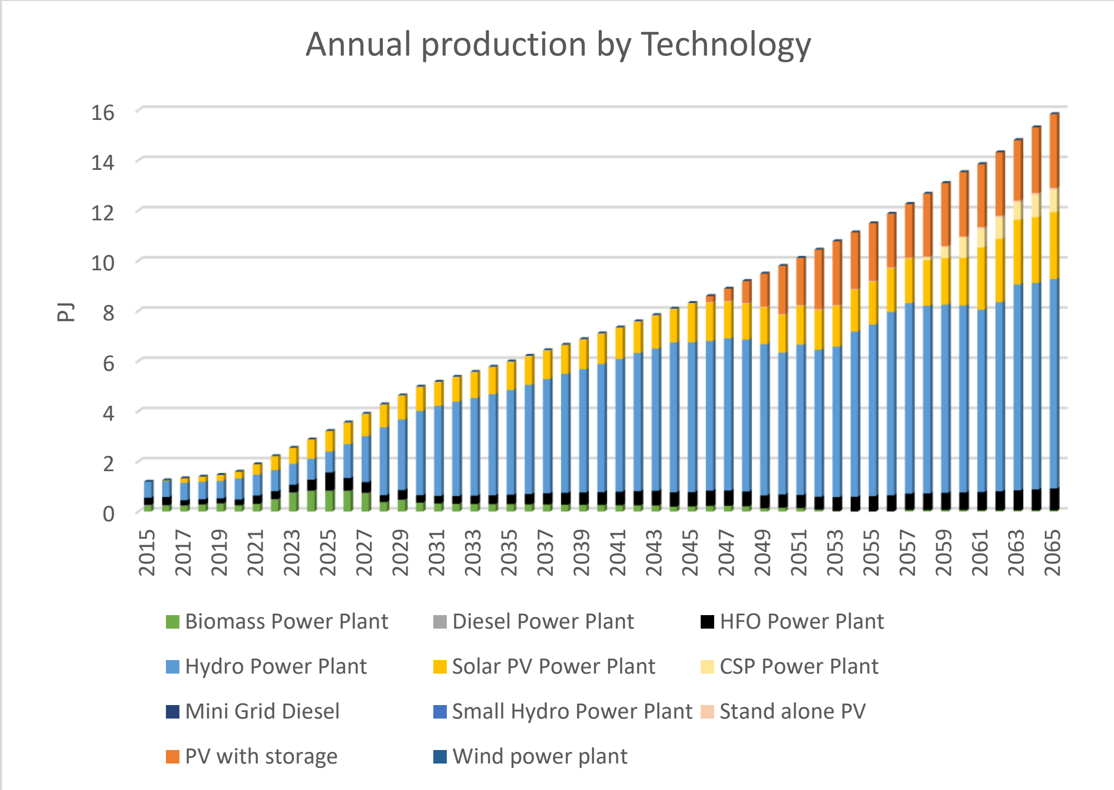


Figure 17. Annual production by technology for the reference scenario

Annual production of electricity is increasing linearly throughout the modelling period. Solar PV and hydropower plant show a stable increase each year following the linear rise in electricity demand. In the whole modelling period, hydro and solar PV power plant produce most of the electricity, with a total contribution of almost 229.42 PJ from the hydro power plant (60% of total production) and 106 PJ (28% of total production) from solar PV. Solar PV with storage does not contribute to electricity production in the beginning. However, it shows a stable increase in electricity production starting from 2046 until 2065 due to lower cost of storage. This technology contributes to a total of 41 PJ of electricity for the whole modelling period. Lastly, HFO and biomass fueled power plant provides a total of 25 PJ and 13 PJ of electricity from 2015 to 2065.

5.2 Medium electricity demand scenario

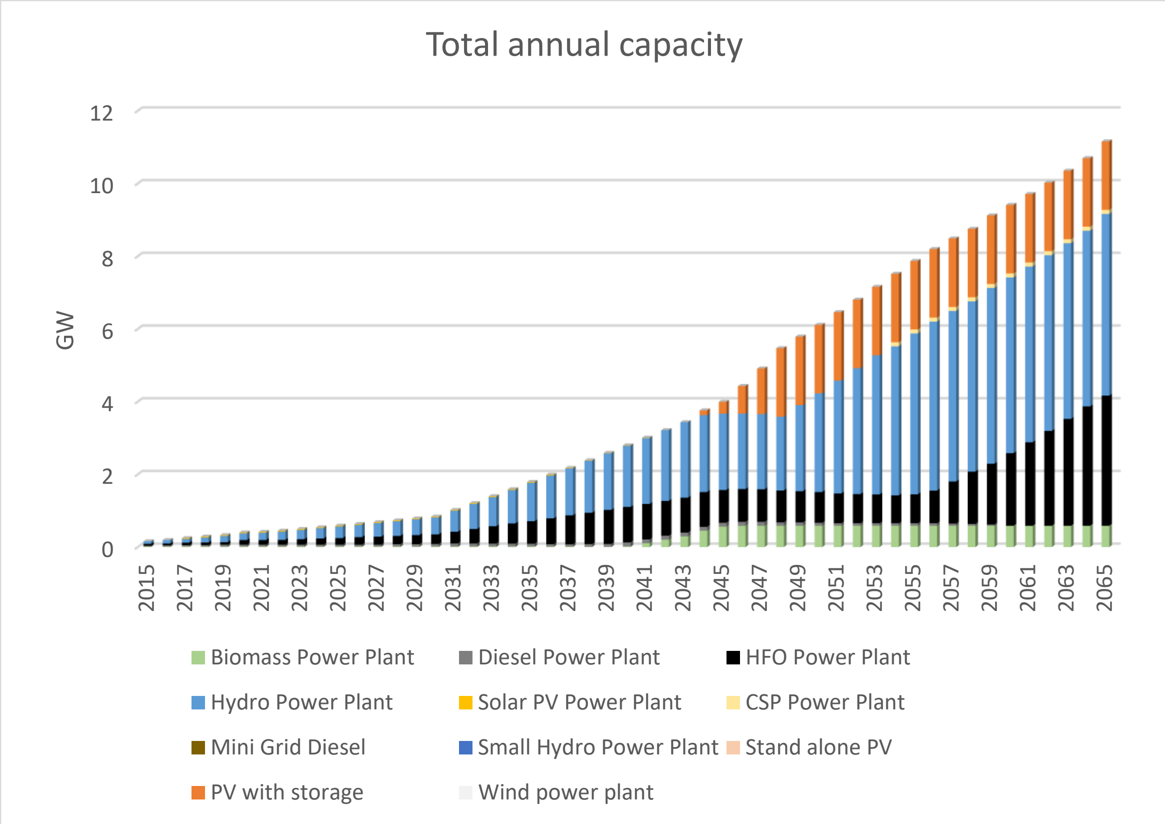


Figure 18. Total annual capacity by technology for medium electricity demand scenario

In the medium electricity demand scenario, the total annual capacity increased steadily from 0.158 GW in 2015 to 0.941 GW in 2030, and 9.06 GW in 2065. The installed capacity is mainly dominated by hydropower plant due to lower electricity generation cost. In 2065, large hydro power plant capacity is almost reaching its maximum available potential capacity, 4.99 GW from 5 GW available. PV with storage enter the market in 2043 with 0,115 GW capacity. In 2050, PV with storage has reached its maximum potential with 1.885 GW.

The capacity of the diesel power plant increases from 0.024 GW in 2015 to 0.083 GW in 2029. Then it decreases gradually to be phased out in 2064. On the other hand, the capacity of biomass and HFO power plant shows a stable increase annually. Biomass power plant capacity was 0.015 GW in 2015, and the capacity increased to 0.572 GW in 2065, while HFO power plant capacity is 0.0523 GW in 2015 and in 2065, the capacity is 1.2 GW.

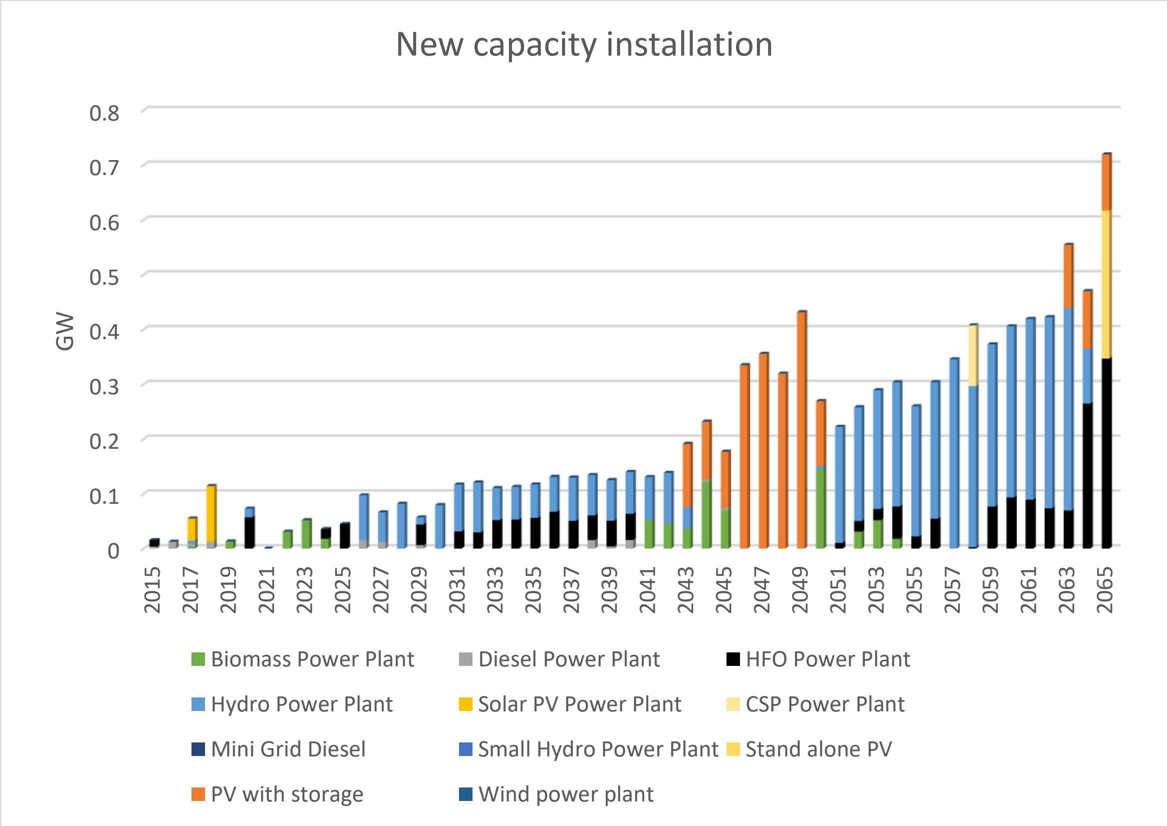


Figure 19. Annual new capacity installation by technology for medium electricity demand scenario

The capacity installation is increasing throughout the year, with an 8.6% annual growth rate from 2015 to 2065. There is a large installation of PV with storage technology in between 2043 and 2050 due to lower cost of storage. Hence in 2050, mini-grid PV has reached its maximum potential capacity. In between 2063 and 2065, there is PV capacity installation again due to the operational life of the technology, which is 20 years. The highest capacity installation happens in 2065 due to high energy demand in that year.

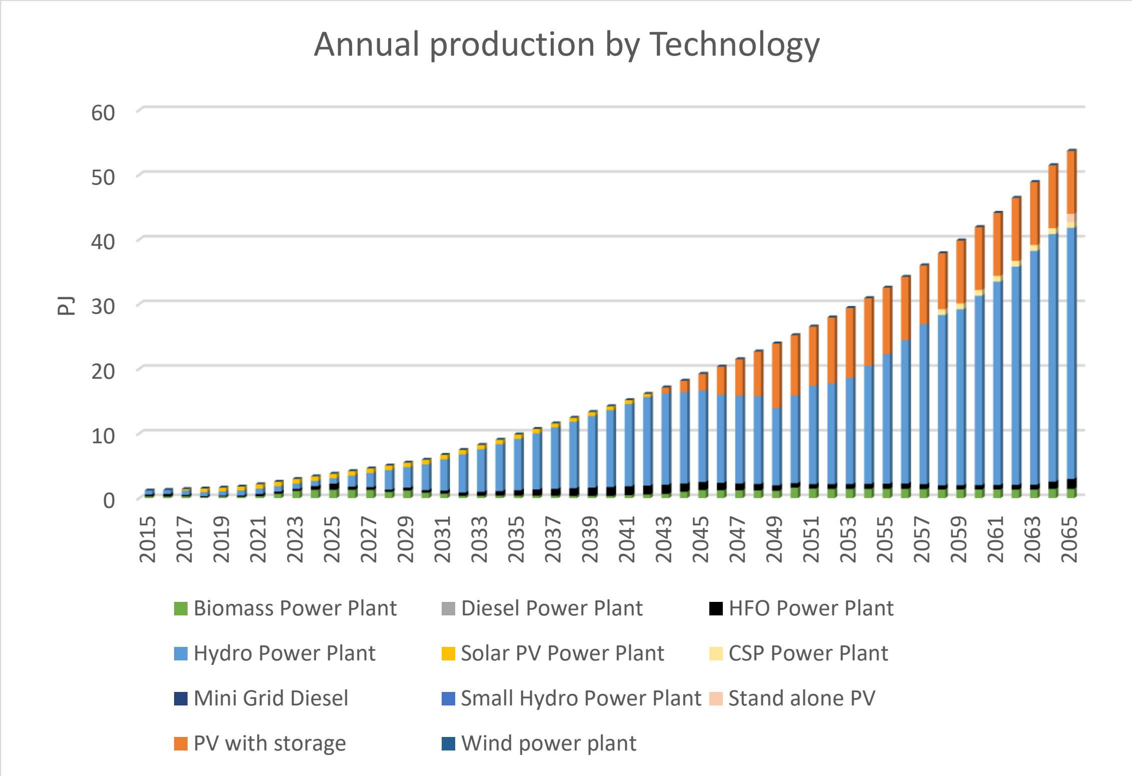


Figure 20. Annual production by technology for medium electricity demand scenario

The total electricity production increases from 1.2 PJ in 2015 to 6 PJ in 2030, and 53.7 PJ in 2065 following the increase in electricity demand. Compared to other technology, hydropower plant shows the most stable contribution to electricity production with an average of 8.6% growth annually. The electricity produced by hydropower plants constitutes approximately 68% of the total generated electricity.

Solar PV does not show significant contribution (less than 0.1 PJ annually) to electricity production from 2015 to 2042. Starting in 2043 till 2065, Solar PV with storage starts to show significant production with an average value of 7.55 PJ in that year due to the lower price of storage. In total, solar PV contributes to 21% of total electricity production in Sierra Leone. Stand-alone PV technology starts to contribute to electricity production in 2064 with 0.23 PJ, and in 2065 the electricity produced is increased significantly with a value of 1.57 PJ. Other technology such as wind turbine and mini-grid diesel does not contribute to any electricity production.

5.3 High electricity demand scenario

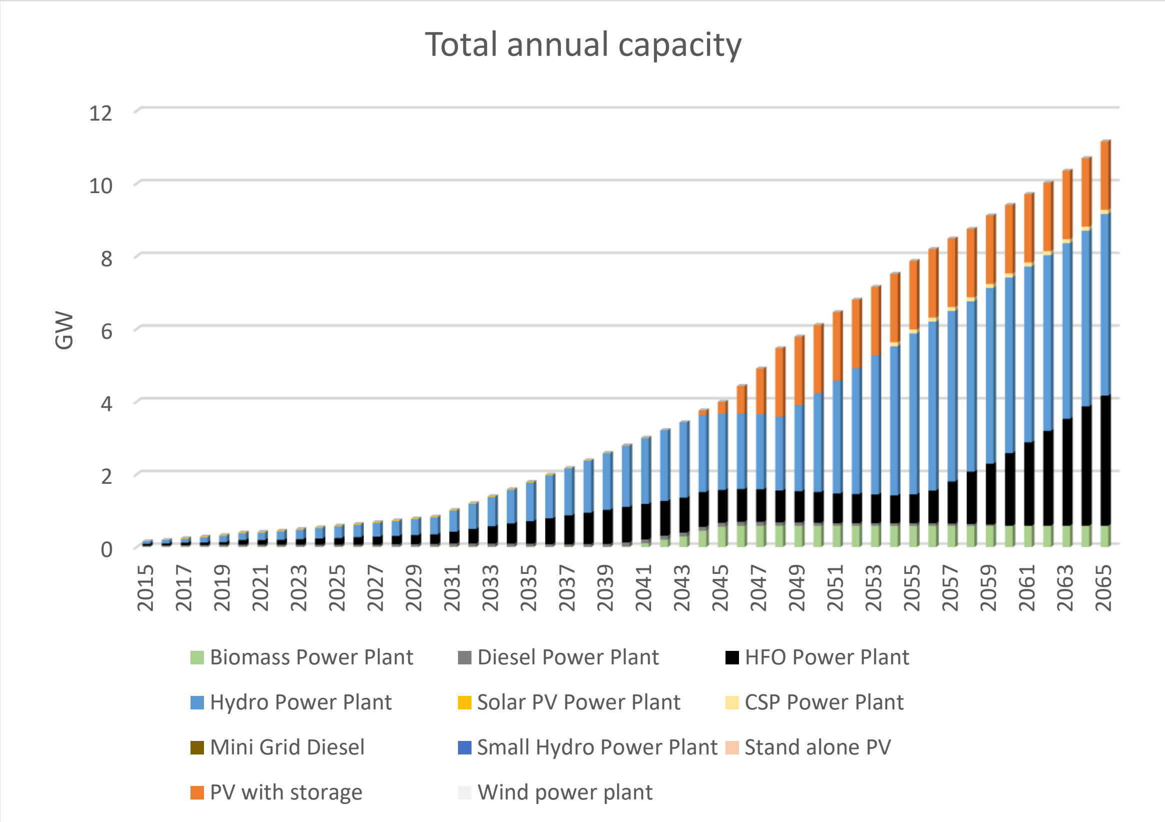


Figure 21. Total annual capacity by technology for high electricity demand scenario

In the high electricity demand scenario, the electricity demand is increasing with a relatively high annual growth rate of 8.6%. Higher capacity installation is needed to fulfill the demand. The capacity for the biomass power plant is continually increasing from 0.015 GW in 2015 to become stable at its maximum potential capacity (0.587 GW) in 2045. HFO power plant shows a similar trend in growth. The capacity of the HFO power plant is only 0.0523 GW in 2015, and it raises to 5 GW in 2065, which is higher than the capacity of hydro (4.6 GW) in the same year. Meanwhile, the capacity of the diesel power plant is increased from 0.2 GW in 2015 to 0.108 GW in 2040, but it is phased out in 2060.

Hydropower plants capacity in 2015 was only 0.07 GW. It increases with an average growth rate of 8.7 % annually until it almost reaches its maximum potential capacity (4.99 GW from 5 GW available potential) in 2065. Mini-grid solar PV technology capacity is 0.05 GW in 2017. Then it decreases slowly until it is phased out in 2042. This technology is shifted by PV with storage technology due to the lower cost of storage. PV with storage technology capacity is built starting from 2036, and the value increased to its maximum potential (1.885 GW) in 2044. CSP technology capacity is zero from 2015 until 2046, and in 2047, it has 0.111 GW capacity, then the capacity value is constant until 2065.

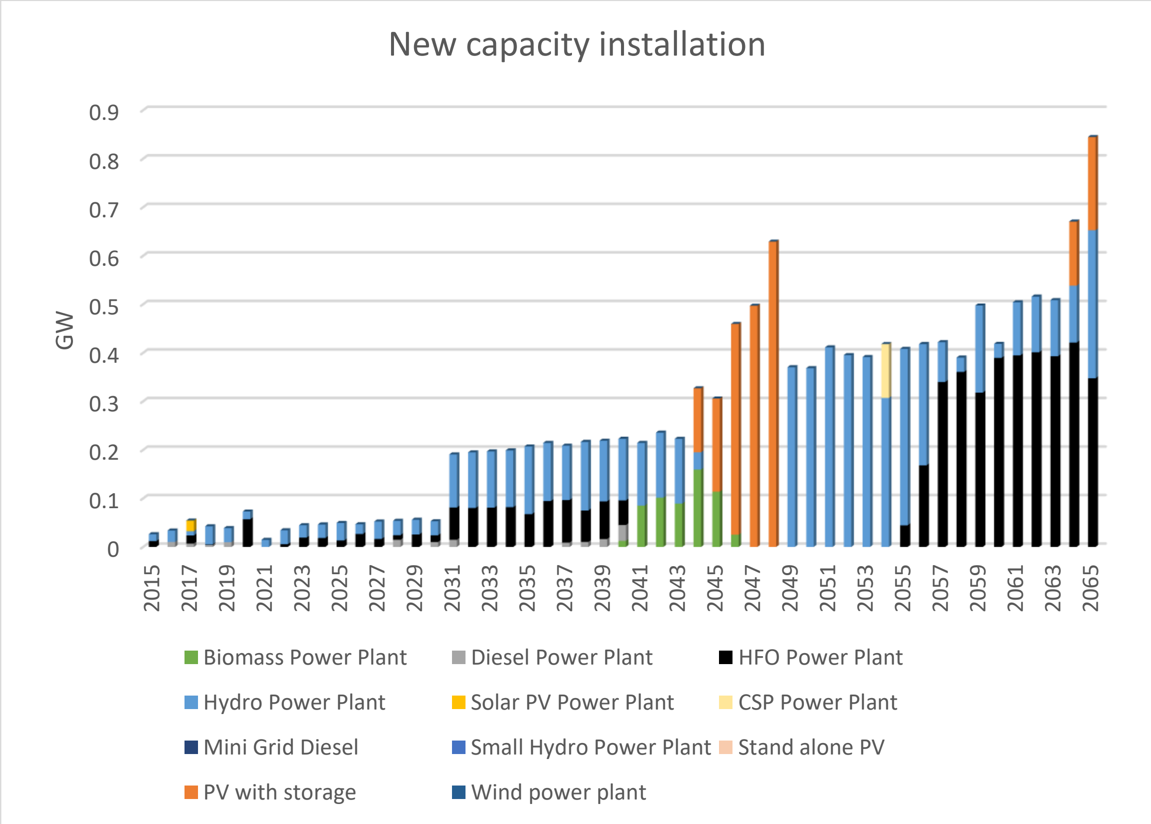


Figure 22. Annual capacity installation by technology for high electricity demand scenario

There is a large installation of PV with storage technology capacity between 2041 until 2044 with a total value of 1.5 GW due to the lower cost of storage technology. The highest capacity installation happens in 2065 with 0.844 GW capacity installation for the particular year due to the massive installation of HFO and hydro technology. Hydro and HFO technology show a similar trend in terms of annual capacity installation. Both technologies show large installations almost every year. In average, there is 0.09 GW installed capacity for HFO technology and 0.117 GW installed capacity for hydro technology annually.

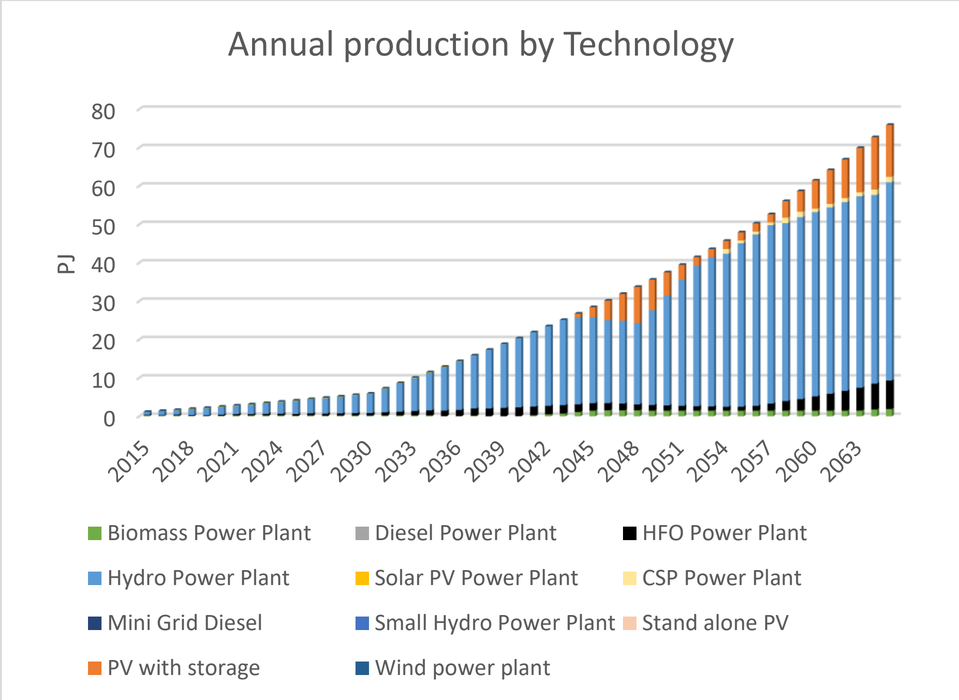


Figure 23. Annual production by technology for high electricity demand scenario

In 2015, the amount of electricity being produced was around 1.2 PJ, while in 2030, approximately 5.95 PJ is being produced, and in 2065, 75.8 PJ electricity is produced. In 2015, hydro technology contributed to around 66% of the electricity production with 0.789 PJ. In 2065, hydro technology still dominates the electricity production with 51.65 PJ (68% from total produced electricity).

Other technology such as biomass, HFO, and solar PV shows a similar growth with a stable increase in electricity production annually. Both biomass and HFO have started to generate electricity since 2015 with just 0.22 PJ and 0.14 PJ to 3.33 PJ and 7.42 PJ in the year 2065. PV technology starts to generate electricity from 2017 with 0.26 PJ and increase gradually to 13.5 PJ in 2065. CSP technology just started producing electricity in 2054 with a value of 1.2PJ. Then the value is constant with slight fluctuation until reaching 1.4 PJ in 2065. Meanwhile, the electricity produced from the diesel power plants is decreased slowly but steadily from 0.031 PJ in 2015 to 0.001892 PJ in 2040.

5.4 Scenario Comparison

In this section, a comparison of the scenarios examined in this study is described.

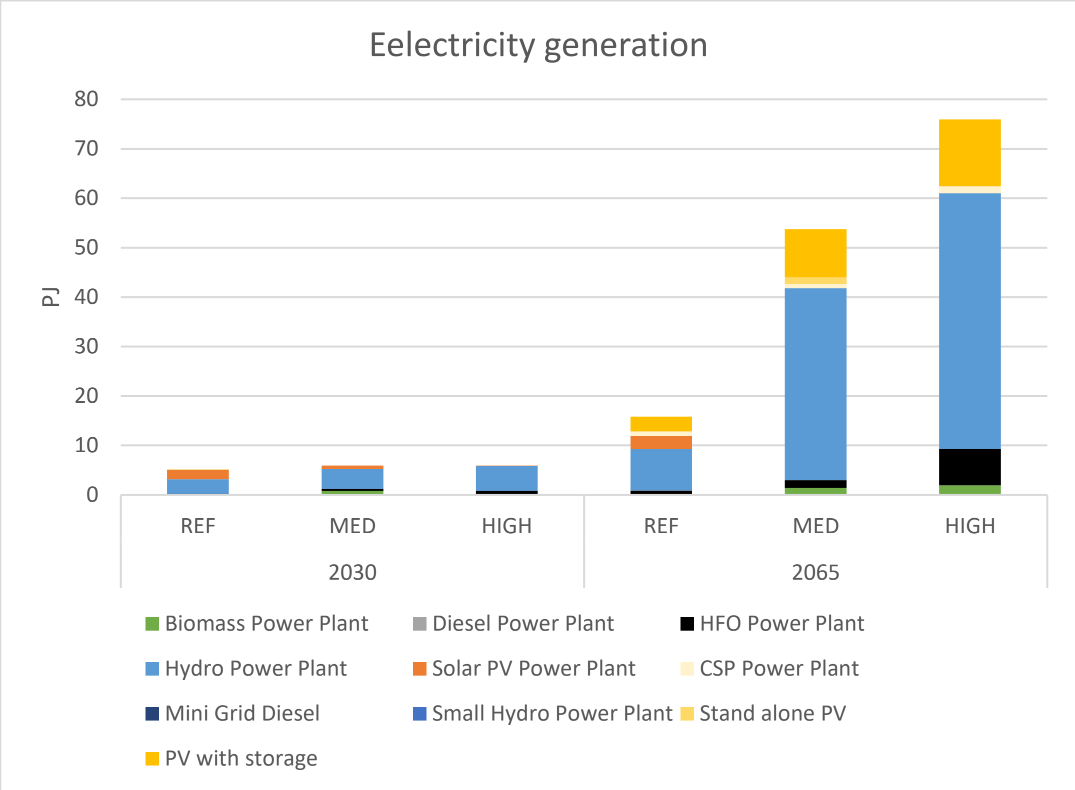


Figure 24. Comparison of annual electricity generation by each technology for all scenarios

In 2015, the total electricity generation was around 1.2 PJ with 15.2% electrification rate. In the reference scenario, the electricity increases to 4.97 PJ in 2030, due to the 92% electrification rate target and 5.8% other sector´s growth rate. It is contrast to the medium and high electricity demand scenarios where electricity generation is increased to around 5.9 PJ in 2030, mainly due to the same electrification rate target (92%) and 7.8% other´s sector growth rate. In 2065, the total electricity generated for the reference scenario is only 16.2 PJ. In the medium and high electricity demand scenarios, the generated electricity is 53.6 PJ and 75.8 PJ correspondingly in 2065. The huge increase in electricity generation in medium and high electricity demand happens because of the increase in average household electricity consumption. Electricity production for all scenarios is dominated by hydro due to its abundant resource and low price of electricity generation. In the reference scenario, besides the high penetration of hydropower plant, PV also gives a high contribution in electricity production (28% of total electricity production) compared to medium (22%) and high (10%) electricity demand scenario. This happens due to the higher penetration of mini-grid PV and stand-alone PV for urban and rural households.

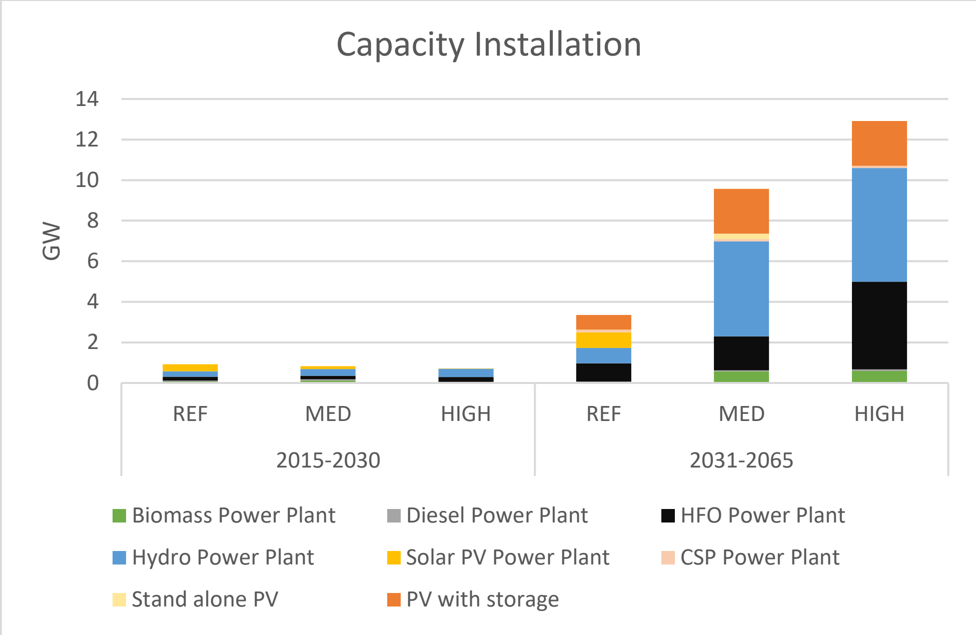


Figure 25. Comparison of capacity investment for all scenarios

Between 2015 and 2030, there is 0.92 GW new installed capacity in the reference scenario. Within the same period, in medium electricity demand scenario, around 0.83 GW capacity is installed, while in high electricity demand scenario 0.72 GW is installed. This result occurs due to high installation of solar PV in the reference scenario which the capacity factor is deficient, so even though the electricity demand is lower in the reference scenario, higher capacity installation is needed. In this period, for medium and high electricity demand scenario, most of the installed capacity is hydropower technology (40% of total installed capacity in medium electricity demand scenario and 56% in high electricity demand scenario). In the reference scenario, most of the installed capacity in this period is solar, with around 36% of total installed capacity.

Between 2031 and 2065, due to the vast difference in the electricity demand, the difference in total installed capacity for all scenarios is significant. In the reference scenario, a total of 3.35 GW capacity is installed, in medium electricity demand scenario 9.57 GW capacity is installed, while in high electricity scenario demand 12.9 GW capacity is being installed. In the reference scenario, solar PV technology dominates the total installed capacity, with a percentage of 45% of the total installation, and the rest of installed capacity is mainly HFO and hydro technology. In medium electricity demand scenario, hydro dominates the capacity installation with a contribution of 49% of total installed capacity. HFO technology shows a significant installation in high electricity demand scenario with a percentage of installation around 38% of total installed capacity besides hydro (43%). All in all, the main similarity in all three scenarios is there is no investment in wind power, small hydropower, and mini-grid diesel technology.

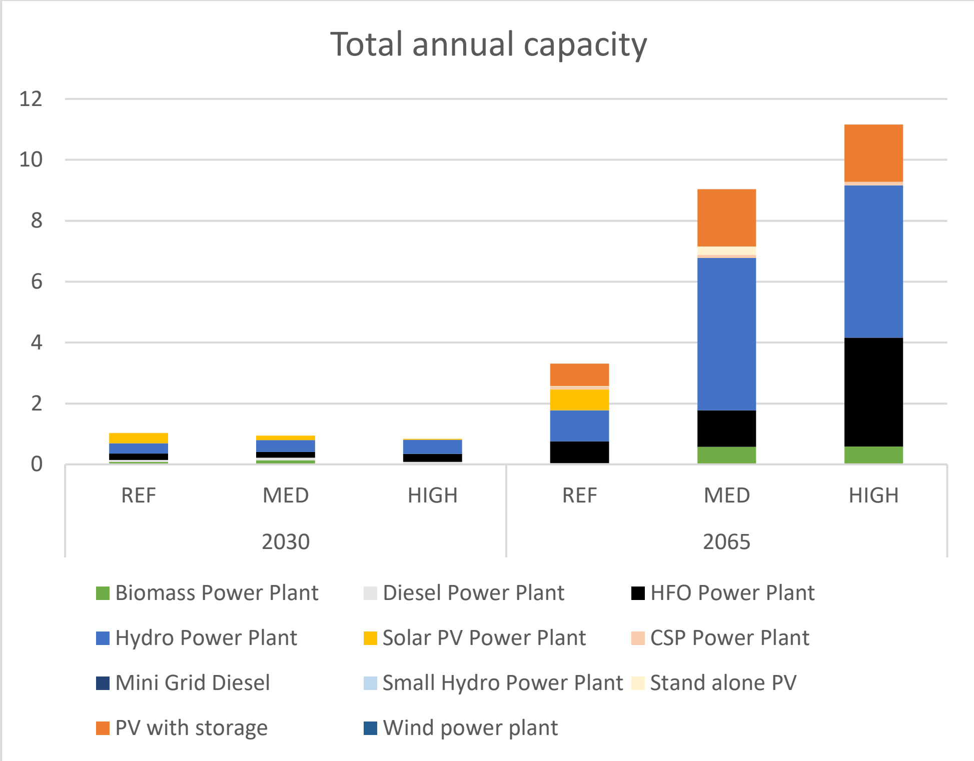


Figure 26. Comparison of total annual installed capacity by each technology for all scenarios

In 2015, the total installed capacity was around 0.158 GW. In the reference scenario, the total installed capacity increases to 1.03 GW in 2030, while in the high and medium electricity demand scenarios, it increases to around 0.83 GW and 0.94 GW correspondingly. Even though the electricity demand in the reference scenario is lower than in the medium electricity demand scenario, the installed capacity is higher in the reference scenario due to the high penetration of mini-grid and stand-alone solar PV, which has a low capacity factor. In 2065, the total annual capacity in reference scenario is increased to 3.3 GW, while in the medium and high electricity demand scenarios, the total installed capacity is 9.03 GW and 11.16 GW.

In 2030, all three scenarios consist of biomass, diesel, and HFO power plant mainly due to remaining capacity from power plant before the modelling period. In the reference scenario, solar PV technology capacity is the highest among other technology (0.596 GW). In medium and high electricity demand scenario, hydro technology has the highest capacity among other technology with 0.388 GW in medium electricity demand scenario and 0.465 GW in high electricity demand scenario.

In 2065, in the high electricity demand, because the capacity of the hydro (5 GW) and solar PV (1.885 GW) power plant is fully utilized, another technology is needed to fulfill the demand. Therefore, in high electricity demand scenario, biomass, HFO, and CSP technology also highly contributes to electricity production. Meanwhile, in the reference and medium electricity demand scenario, solar and hydropower plant dominates capacity available due to lower demand.

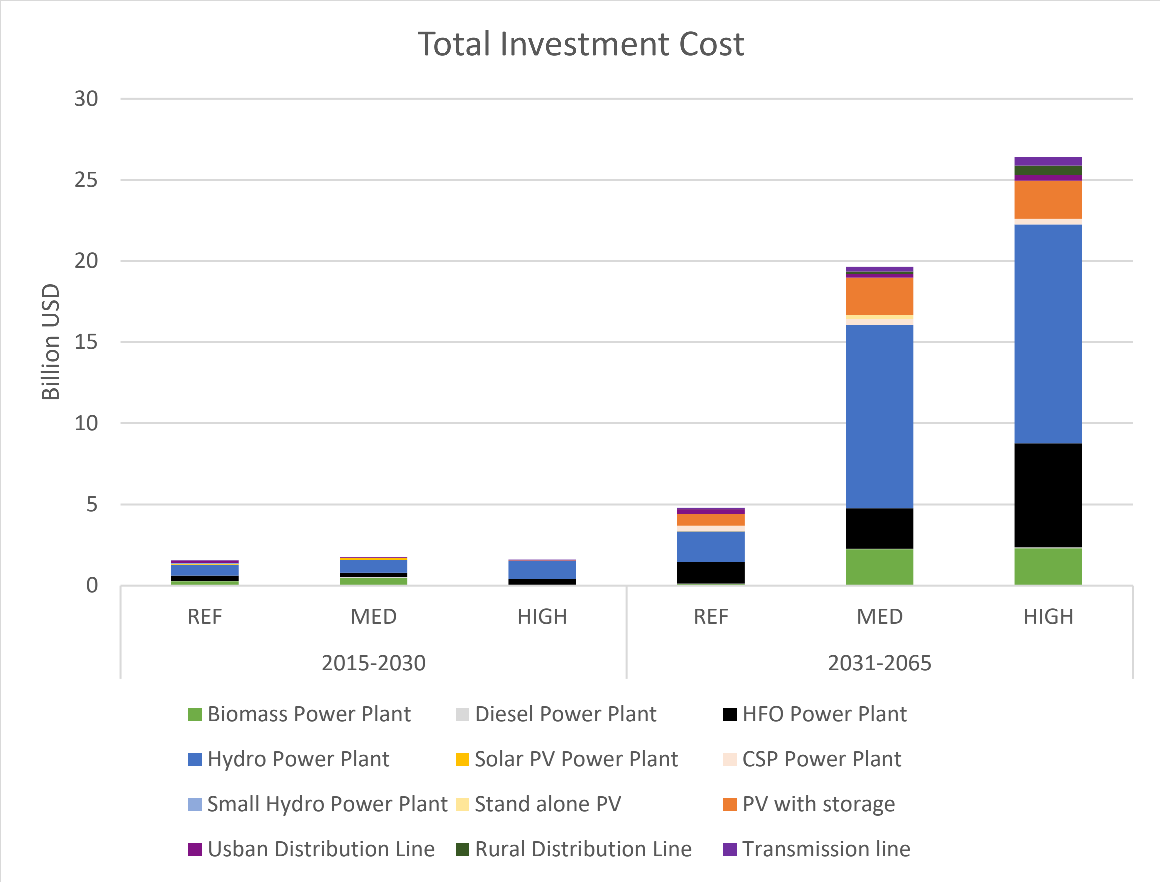


Figure 27. The total investment cost for all scenario (in billion USD)

Between 2015 and 2030, around 1.54 billion USD is invested in covering the electricity demand for the reference scenario, while the medium and high electricity demand scenario require 1.75 billion USD and 1.6 billion USD. Between 2031-2065, in the medium and high electricity demand scenario, around 19.6 billion USD and 26.4 billion USD are needed for the investment, while, only 4.8 billion USD is needed in the reference scenario. In all scenario, investment capital mainly goes to the development of technology, while only a few proportions go to the transmission and distribution line extension. In the reference scenario, only 8.4% of the investment goes to transmission and distribution line extension, while in the medium and high electricity scenario, around 3.3% and 5.3% of the investment goes to grid extension.

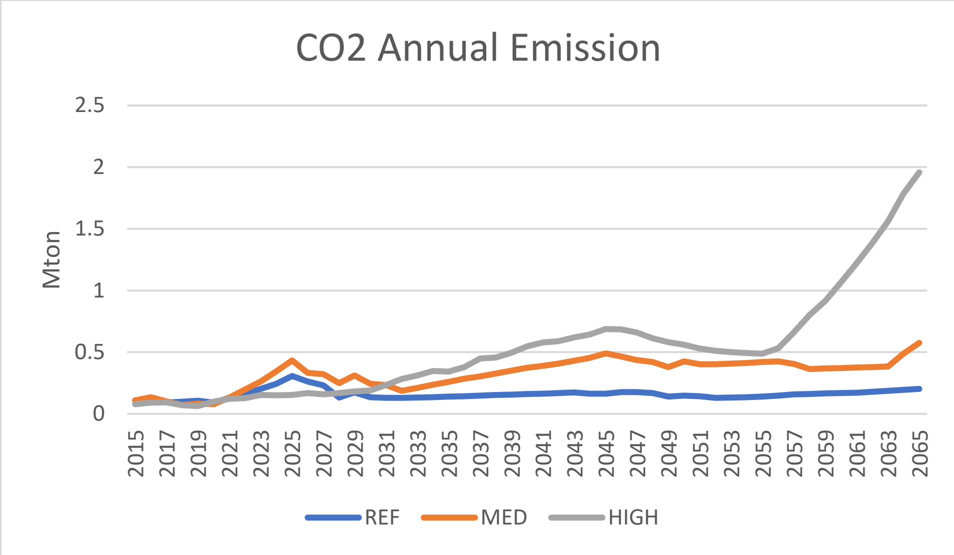


Figure 28. Comparison of annual CO2 emission for all scenarios

The CO_2 emissions are increasing over time for all scenarios due to the gradual increase of the installed capacity of HFO and biomass power plant. The CO_2 emissions were only 0.1 Mton in 2015, while by 2065, in the reference scenario, they increase approximately by 2 times (0.2 Mton). In the medium electricity demand scenario, the emissions increase approximately by 6 times by 2065 due to higher installed capacity of biomass and HFO plants compared to the reference scenario. In the high electricity demand scenario, the emissions increase to almost 25 times compared to 2015 levels due to the much higher installation of biomass and HFO power plants.

6. Conclusions

Three scenarios, listed as reference, medium electricity demand, and high electricity demand scenarios, are analyzed in this study. The difference between the three scenarios is in the electricity demand, the total electricity demand was 1.196 PJ in 2015. In 2030, the electricity demand for the reference scenario is 5.13 PJ while in the medium and high electricity demand scenario increases to 6 PJ. In 2065, the electricity demand in the reference scenario is 16.2 PJ while in the medium electricity demand scenario is 54.24 PJ, and in the high electricity demand scenario Is 75.8 PJ.

In 2015, the installed capacity was around 0.144 GW. By 2030, the capacity increases to 1.03 GW for the reference scenario and around 0.94 GW for medium and around 8.34 GW for high electricity demand scenarios. Even though the electricity demand is higher in the medium and high electricity demand scenarios, the capacity in the reference scenario is higher due to the massive installation of Solar PV technology which has a low capacity factor. In 2065, the capacity is increased to 3.3 GW for the reference scenario. In the medium and high electricity demand scenarios, the capacity is increased to 9 GW and 11.16 GW in the same year.

Every scenario suggests building a large hydropower plant due to the country´s high hydro potential as well as the fact this type of power generation technology can provide cheap electricity. In the reference scenario, where OnSSET analysis is included, there is also a considerable penetration of solar PV technologies. A total of 338 MW capacity of mini-grid solar PV stand-alone PV is installed to fulfill the electricity demand in 2030, compared to no PV technology in 2015. Then, in 2065, a total of 1.525 GW additional capacity of all PV technologies is installed to fulfill the electricity demand. The highest contributor to PV technology is mini-grid solar PV with 1.84 GW capacity. Therefore, in the reference scenario, mini-grid PV power plant contributes to 45.6% of total electricity production in 2065. However, hydropower plant still produces most of the electricity, which is around 52.7%. In the medium electricity demand scenario, hydropower plant contributes to 72% of the electricity production, while in the high electricity demand scenario, hydro power plant contributes to around 68% of the electricity being produced in 2065.

The CO_2 emissions based on the power sector are low for both reference and medium electricity demand scenarios due to the low production from fossil-fueled and biomass power plants. However, in high electricity demand scenario, the CO_2 emissions are much higher due to higher electricity production from biomass and HFO technology. In the reference scenario, the CO_2 emissions increase from 0.11 Mton in 2015 to 0.2 Mton in 2065. Showing a similar trend, in the medium electricity demand scenario, CO_2 emissions increase to 0.575 Mton in 2065. The high electricity demand scenario shows the utmost increase in CO_2 emissions compared to the other two scenarios with almost 2 Mton CO_2 in 2065.

In the Reference scenario, Sierra Leone should invest in a total of 6.34 billion USD in its power sector in order to cover its electricity needs the period 2015-2065. In the medium electricity demand scenario, a total of 21.4 billion USD will be required to be invested. Meanwhile, in the high electricity demand scenario, 28 billion USD will be needed. Hydro technology contributes to most of the investment costs. Around 40% of the total investment costs are spent on this technology in the reference scenario, while for the medium and high electricity demand scenarios, the percentage is 56.4% and 52%. In addition, future investments in biomass and HFO power plants also contribute at a high level to the high investment cost for all scenarios.

In conclusion, as indicated in the results of this study the country could exploit its high potential of renewable energy, especially in hydro (5 GW) and solar (1.885 GW) to cover its future electricity needs.

7. Limitations-Future Work

In this section, the limitation of this study as well as further possible improvements are discussed. The main limitation is the availability of data in order to develop the model. Different sources provided quite different data about the country´s power system. However, since the model is open source, this study can be updated.

Besides, it has been assumed that economic growth in the country will remain constant throughout the modelling period. However, this can be improved in the future by using better forecasting technique such as multiple linear regression. Electricity consumption per household is also expected to be persistent. Moreover, since the results show that the electricity produced is mainly based on hydro technology, the capacity factor and availability factor value of this technology will profoundly affect the investment result.

Future work can be improved by doing a sensitivity analysis regarding techno-economic costs, fuel prices to identify what could be the significant inputs to improve the analysis. Moreover, a more detailed analysis of the electricity trade should be examined. Additionally, considering the poor transmission and distribution technology now, analysis of efficiency improvements should be considered. In addition, reserve margin and storage should be implemented in the model to capture the intermittency of renewable energy technologies. Furthermore, since the focus of this study is mainly different electricity demand projections, it will be useful to examine consumer´s behaviour and how it affects the electricity consumption per household.

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Appendix A: Detailed techno-economic parameter

Table 19. Techno-economic parameter for each power plant (IRENA, 2018)

Technology	Capital Cost (USD/kW)	Fixed Cost (USD/kW)	Variable Cost (USD/kWh)	Life (Years)	Efficiency	Availability	Construction time (Years)	Capacity Factor (average)
BIOMASS 1	3917	20.4	5.76	30	0.38	0.86	4	0.56
BIOMASS 2	3917	20.4	5.76	30	0.38	0.86	4	0.56
DIESEL MAKENI	1240	36.8	51.52	25	0.35	0.8	2	0.56
DIESEL OHF	1240	36.8	51.52	30	0.35	0.8	2	0.56
DIESEL ODS	1240	36.8	51.52	30	0.35	0.8	2	0.56
HFO BKPS	1480	44	23.76	25	0.35	0.8	2	0.56
HFO BLACKHALL	1480	44	23.76	25	0.35	0.8	2	0.56
HFO KINGTOM	1480	44	23.76	25	0.35	0.8	2	0.56
HFO KOIDU	1480	44	23.76	25	0.35	0.8	2	0.56
HFO KONO	1480	44	23.76	25	0.35	0.8	2	0.56
HFO LUNGI	1480	44	23.76	25	0.35	0.8	2	0.56
HFO LUNSAR	1480	44	23.76	25	0.35	0.8	2	0.56
HFO WAPP	2208	44	23.76	25	0.35	0.8	2	0.56
HYDRO BANKASOKA	4045	124	0	30	-	0.59	2	0.54
HYDRO BENKONGOR1	2660.00	79.8	0	50	-	0.78	4	0.45
HYDRO BENKONGOR2	2660.00	79.8	0	50	-	0.59	4	0.54
HYDRO BENKONGOR3	2660.00	79.8	0	50	-	0.56	4	0.54
HYDRO BUMBUNA1		66.24	0	50	-	0.66	4	0.555
HYDRO BUMBUNA2	2120.00	63.59	0	50	-	0.63	4	0.544
HYDRO BUMBUNA3	2120.00	63.59	0	50	-	0.5	4	0.54
HYDRO BUMBUNA45	2120.00	63.59	0	50	-	0.59	4	0.54

HYDRO CHARLOTTE	4045	124	0	30	-	0.59	2	0.54
HYDRO ENVISAGEE	2784.00	83.51	0	50	-	0.66	4	0.54
HYDRO GOMA1		66.24	0	30	-	0.59	2	0.54
HYDRO MAKALI	4045	124	0	30	-	0.59	2	0.54
HYDRO MOYAMBA	3814	124	0	30	-	0.5	2	0.54
HYDRO YELE	4045	124	0	30	-	0.59	2	0.54
PV NEWTON	1753.92	21.74	0	25	-	1	1	0.18
PV SOLAR2	1840	21.74	0	20	-	0.12	1	0.18
PV YAMANDU	1753.92	21.74	0	25	-	1	1	0.18
EXTRA BIO	3917.00	82	5.76	30	0.38	0.5	4	0.56
EXTRA CSP	4010	65	0	25	-	1	4	0.35
EXTRA CSP + STO	5900	93	0	25	-	1	4	0.4
EXTRA DIESEL	1240.00	35	51.52	25	0.35	0.8	2	0.56
EXTRA DIESEL INDUSTRY	730.00	21	51.52	20	0.35	0.8	0	0.56
EXTRA DIESEL RURAL	750.00	23	51.52	10	0.16	0.3	0	0.56
EXTRA DIESEL URBAN	750.00	23	51.52	10	0.16	0.3	0	0.56
EXTRA HFO	1480.00	44	23.76	25	0.35	0.8	2	0.56
EXTRA HYDRO	2660.00	79.8	0	30	-	0.5	4	0.55
EXTRA HYDROS	4045	124	0	30	-	0.5	2	0.55
EXTRA PV	1753.92	21.74	0	25	-	1	1	0.1548
EXTRA PV STAND ALONE RURAL	3782.04	50	0	20	-	1	1	0.1548
EXTRA PV STAND ALONE URBAN	3782.04	50	0	20	-	1	1	0.1548
EXTRA PV + STORAGE	5010.00	61	0	20	-	1	1	0.255
OFFSHORE WIND	4258.01	159.1364205	0	25	-	1	2	0.068
ONSHORE WIND	1550.16	95	0	25	-	1	2	0.09

Appendix B: Detailed OnSSET recommendation

Table 20. OnSSET recommendation for reference scenario

	Population 2030	New Connection 2030	Capacity 2030 (MW)	Investment 2030 (Milion USD)	Population 2065	New Connection 2065	Capacity 2065 (MW)	Investment 2065 (Milion USD)
Grid	2653643	1541474	40	197.83	7482885	4041123	95	993.94
SA_Diesel	0	0	0	0	25136	11772	0	0.05
SA_PV	206000	206000	5	33.1	149675	70442	1	7.7
MG_Diesel	0	0	0	0	0	0	0	0
MG_PV	6275298	6275298	277	632.33	12036886	6436302	401	359.3
MG_Wind	0	0	0	0	0	0	0	0
MG_Hydro	0	0	0	0	0	0	0	0
Total	9134942	8022773	323	863.27	19694584	10559641	498	1360.99